



PostgreSQL Physical Model

Schema for:

Model name: New model

Author:

Version:

File name: psql_schema_assig2.hck.json

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1. Model

2. Schemas

2.1 psql_scheme

2.1.2. Tables

- 2.1.2.1 clients
- 2.1.2.2 campaigns
- 2.1.2.3 users
- 2.1.2.4 campaigns_ab_test
- 2.1.2.5 campaigns_hour_limit
- 2.1.2.6 campaigns_test
- 2.1.2.7 campaigns_position
- 2.1.2.8 messages
- 2.1.2.9 messages_device_info
- 2.1.2.10 messages_category
- 2.1.2.11 messages_recipient_info
- 2.1.2.12 email_providers
- 2.1.2.13 message_opens
- 2.1.2.14 message_clicks
- 2.1.2.15 message_negatives
- 2.1.2.16 message_purchases
- 2.1.2.17 events
- 2.1.2.18 products
- 2.1.2.19 categories
- 2.1.2.20 friends

3. Relationships

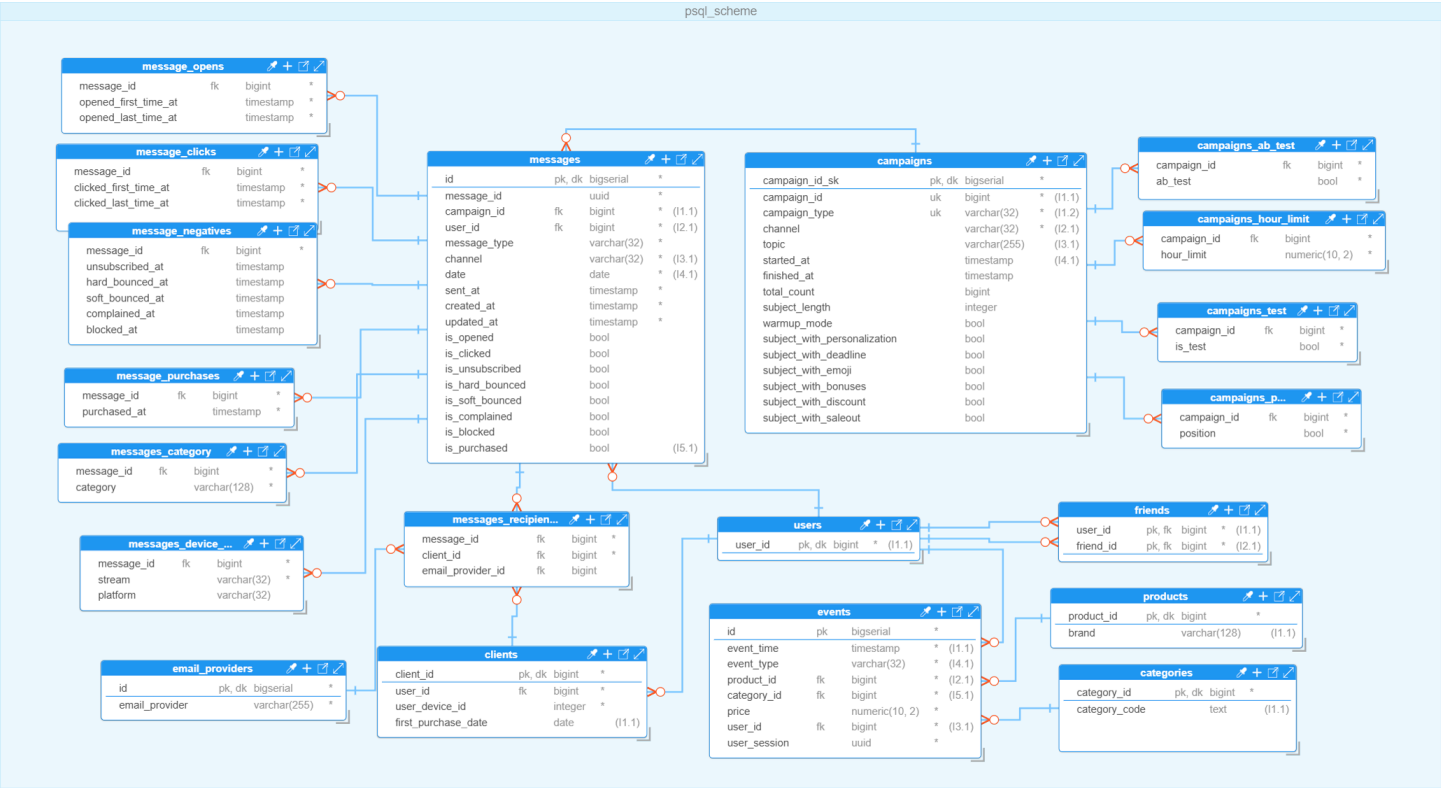
- 3.1 fk_users_user_id_to_clients_user_id
- 3.2 fk_campaigns_id_to_campaigns_ab_test_campaign_id
- 3.3 fk_campaigns_id_to_campaigns_ab_test_campaign_id
- 3.4 fk_campaigns_id_to_campaigns_ab_test_campaign_id
- 3.5 fk_campaigns_id_to_campaigns_ab_test_campaign_id
- 3.6 fk_campaigns_campaign_id_sk_to_messages_campaign_id
- 3.7 fk_users_user_id_to_messages_user_id
- 3.8 fk_messages_id_to_message_device_info_message_id
- 3.9 fk_messages_id_to_messages_category_message_id
- 3.10 fk_messages_id_to_messages_recipient_info_message_id
- 3.11 fk_clients_client_id_to_messages_recipient_info_client_id
- 3.12 fk_email_providers_id_to_messages_recipient_info_email_provider_id
- 3.13 fk_messages_id_to_message_opens_message_id

- 3.14 fk_messages_id_to_message_opens_message_id
- 3.15 fk_messages_id_to_message_opens_message_id
- 3.16 fk_messages_id_to_message_opens_message_id
- 3.17 fk_users_user_id_to_events_user_id
- 3.18 fk_products_product_id_to_events_product_id
- 3.19 fk_users_user_id_to_friends_user_id
- 3.20 fk_users_user_id_to_friends_friend_id
- 3.21 fk_categories_category_id_to_events_category_id

1. MODEL

1.1 Model New model

1.1.1 New model Entity Relationship Diagram



1.1.2 New model Properties

1.1.2.1 Details tab

PROPERTY	VALUE
Model name	New model
DB vendor	PostgreSQL
DB version	v16.x
Tablespace	pg_default
Encoding	UTF8
Custom scripts	
Lineage capture	false

1.1.2.2 Options tab

PROPERTY	VALUE
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1.1.3 New model User-Defined Types

2. SCHEMAS

2.1 Schema **psql_scheme**

2.1.1 **psql_scheme** Properties

PROPERTY	VALUE
Schema name	psql_scheme
Activated	true
If not exist	true

2.1.2 **psql_scheme** Tables

2.1.2.1 Table **clients**

2.1.2.1.1 **clients** Properties

PROPERTY	VALUE
Table	clients
Technical name	clients
Activated	true
Schema	psql_scheme
Additional properties	false
If not exists	true
Tablespace	pg_default

2.1.2.1.2 **clients** Column

COLUMN	TYPE	REQ	KEY	DESCRIPTION	COMMENTS
client_id	bigint	true	pk, dk		
user_id	bigint	true	fk		
user_device_id	integer	true			
first_purchase_date	date	false			

2.1.2.1.2.1 Column **client_id**

2.1.2.1.2.1.1 **client_id** properties

PROPERTY	VALUE
Business Name	client_id
Technical name	client_id
Activated	true
Type	numeric
Subtype	bigint
Generated column	true
Not null	true
Primary key	true
Excl min	false
Excl max	false

2.1.2.1.2.2 Column **user_id**

2.1.2.1.2.2.1 user_id properties

PROPERTY	VALUE
Business Name	user_id
Technical name	user_id
Activated	true
Type	numeric
Subtype	bigint
Not null	true
Primary key	false
Unique	false
Foreign table	users
Foreign field	user_id
Relationship type	Foreign Key
Excl min	false
Excl max	false

2.1.2.1.2.3 Column user_device_id

2.1.2.1.2.3.1 user_device_id properties

PROPERTY	VALUE
Business Name	user_device_id
Technical name	user_device_id
Activated	true
Type	numeric
Subtype	integer
Not null	true
Primary key	false
Unique	false
Excl min	false
Excl max	false

2.1.2.1.2.4 Column first_purchase_date

2.1.2.1.2.4.1 first_purchase_date properties

PROPERTY	VALUE
Business Name	first_purchase_date
Technical name	first_purchase_date
Activated	true
Type	datetime
Subtype	date
Not null	false
Primary key	false
Unique	false

2.1.2.1.3 clients Composite keys

PROPERTY	VALUE
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2.1.2.1.4 clients Indexes

2.1.2.1.4.1 Index

PROPERTY	
Name	user
Activated	true
Method	btree
Unique	
Concurrent build	
If not exist	true
Only	true
Columns	
	first_purchase_date
Expressions	
[1] value	
Include non-key columns	
With storage parameters	
[1] Fill factor	
Deduplicate items	
Tablespace	pg_default
Where constraint	
Comment	

2.1.2.1.5 clients Target Script

```
-- Generated by Hackolade Studio on: 2026-03-14T21:03:03+03:00
-- Application version: 8.9.6
-- PostgreSQL plugin version: 0.2.43
-- Custom scripts used: None

CREATE SCHEMA IF NOT EXISTS psql_scheme;
SET search_path TO psql_scheme, public;

CREATE TABLE IF NOT EXISTS psql_scheme.clients (
  client_id bigint PRIMARY KEY NOT NULL,
  user_id bigint NOT NULL,
  user_device_id integer NOT NULL,
  first_purchase_date date,
  CONSTRAINT fk_users_user_id_to_clients_user_id FOREIGN KEY (user_id) REFERENCES psql_scheme.users (user_id)
) TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS "user"
ON ONLY psql_scheme.clients USING BTREE (first_purchase_date ASC)
TABLESPACE pg_default;
```

2.1.2.2 Table campaigns

2.1.2.2.1 campaigns Properties

PROPERTY	VALUE
Table	campaigns
Technical name	campaigns
Activated	true
Schema	psql_scheme
Additional properties	false
If not exists	true
Tablespace	pg_default

2.1.2.2.2 campaigns Column

COLUMN	TYPE	REQ	KEY	DESCRIPTION	COMMENTS
campaign_id_sk	bigserial	true	pk, dk		
campaign_id	bigint	true	uk		
campaign_type	varchar(32)	true	uk		
channel	varchar(32)	true			
topic	varchar(255)	false			
started_at	timestamp	false			
finished_at	timestamp	false			
total_count	bigint	false			
subject_length	integer	false			
warmup_mode	boolean	false			
subject_with_personalization	boolean	false			
subject_with_deadline	boolean	false			
subject_with_emoji	boolean	false			
subject_with_bonuses	boolean	false			
subject_with_discount	boolean	false			
subject_with_saleout	boolean	false			

2.1.2.2.2.1 Column campaign_id_sk

2.1.2.2.2.1.1 campaign_id_sk properties

PROPERTY	VALUE
Business Name	campaign_id_sk
Technical name	campaign_id_sk
Activated	true
Type	numeric
Subtype	bigserial
Generated column	true
Not null	true
Primary key	true
Excl min	false
Excl max	false

2.1.2.2.2.2 Column campaign_id

2.1.2.2.2.1 campaign_id properties

PROPERTY	VALUE
Business Name	campaign_id
Technical name	campaign_id
Activated	true
Type	numeric
Subtype	bigint
Not null	true
Primary key	false
Unique	true
Alternate Key	false
Excl min	false
Excl max	false

2.1.2.2.2.3 Column campaign_type**2.1.2.2.2.3.1 campaign_type properties**

PROPERTY	VALUE
Business Name	campaign_type
Technical name	campaign_type
Activated	true
Type	char
Subtype	varchar
Length	32
Not null	true
Primary key	false
Unique	true
Alternate Key	false

2.1.2.2.2.4 Column channel**2.1.2.2.2.4.1 channel properties**

PROPERTY	VALUE
Business Name	channel
Technical name	channel
Activated	true
Type	char
Subtype	varchar
Length	32
Not null	true
Primary key	false
Unique	false

2.1.2.2.2.5 Column topic

2.1.2.2.2.5.1 topic properties

PROPERTY	VALUE
Business Name	topic
Technical name	topic
Activated	true
Type	char
Subtype	varchar
Length	255
Not null	false
Primary key	false
Unique	false

2.1.2.2.2.6 Column started_at**2.1.2.2.2.6.1 started_at properties**

PROPERTY	VALUE
Business Name	started_at
Technical name	started_at
Activated	true
Type	datetime
Subtype	timestamp
Not null	false
Primary key	false
Unique	false

2.1.2.2.2.7 Column finished_at**2.1.2.2.2.7.1 finished_at properties**

PROPERTY	VALUE
Business Name	finished_at
Technical name	finished_at
Activated	true
Type	datetime
Subtype	timestamp
Not null	false
Primary key	false
Unique	false

2.1.2.2.2.8 Column total_count

2.1.2.2.8.1 total_count properties

PROPERTY	VALUE
Business Name	total_count
Technical name	total_count
Activated	true
Type	numeric
Subtype	bigint
Not null	false
Primary key	false
Unique	false
Excl min	false
Excl max	false

2.1.2.2.9 Column subject_length**2.1.2.2.9.1 subject_length properties**

PROPERTY	VALUE
Business Name	subject_length
Technical name	subject_length
Activated	true
Type	numeric
Subtype	integer
Not null	false
Primary key	false
Unique	false
Excl min	false
Excl max	false

2.1.2.2.10 Column warmup_mode**2.1.2.2.10.1 warmup_mode properties**

PROPERTY	VALUE
Business Name	warmup_mode
Technical name	warmup_mode
Activated	true
Type	boolean
Not null	false
Default	false

2.1.2.2.11 Column subject_with_personalization**2.1.2.2.11.1 subject_with_personalization properties**

PROPERTY	VALUE
Business Name	subject_with_personalization
Technical name	subject_with_personalization
Activated	true
Type	boolean
Not null	false
Default	false

2.1.2.2.12 Column subject_with_deadline

2.1.2.2.2.12.1 **subject_with_deadline** properties

PROPERTY	VALUE
Business Name	subject_with_deadline
Technical name	subject_with_deadline
Activated	true
Type	boolean
Not null	false
Default	false

2.1.2.2.2.13 Column **subject_with_emoji**

2.1.2.2.2.13.1 **subject_with_emoji** properties

PROPERTY	VALUE
Business Name	subject_with_emoji
Technical name	subject_with_emoji
Activated	true
Type	boolean
Not null	false
Default	false

2.1.2.2.2.14 Column **subject_with_bonuses**

2.1.2.2.2.14.1 **subject_with_bonuses** properties

PROPERTY	VALUE
Business Name	subject_with_bonuses
Technical name	subject_with_bonuses
Activated	true
Type	boolean
Not null	false
Default	false

2.1.2.2.2.15 Column **subject_with_discount**

2.1.2.2.2.15.1 **subject_with_discount** properties

PROPERTY	VALUE
Business Name	subject_with_discount
Technical name	subject_with_discount
Activated	true
Type	boolean
Not null	false
Default	false

2.1.2.2.2.16 Column **subject_with_saleout**

2.1.2.2.2.16.1 subject_with_saleout properties

PROPERTY	VALUE
Business Name	subject_with_saleout
Technical name	subject_with_saleout
Activated	true
Type	boolean
Not null	false
Default	false

2.1.2.2.3 campaigns Composite keys

PROPERTY	VALUE
Unique key	
[1] Constraint name	campaigns_key
Key	
	campaign_id
	campaign_type
Include non-key columns	
With storage parameters	
Index tablespace	
Comment	
Deferrable	
Alternate key	

2.1.2.2.4 campaigns Indexes

2.1.2.2.4.1 Index

PROPERTY	
Name	cmp_id
Activated	true
Method	btree
Unique	
Concurrent build	
If not exist	true
Only	true
Columns	
	campaign_id
	campaign_type
Expressions	
[1] value	
Include non-key columns	
With storage parameters	
[1] Fill factor	
Deduplicate items	
Tablespace	pg_default
Where constraint	
Comment	

2.1.2.2.4.2 Index

PROPERTY	
Name	cmp_channel
Activated	true
Method	btree
Unique	
Concurrent build	
If not exist	true
Only	true
Columns	
	channel
Expressions	
[1] value	
Include non-key columns	
With storage parameters	
[1] Fill factor	
Deduplicate items	
Tablespace	pg_default
Where constraint	
Comment	

2.1.2.2.4.3 Index

PROPERTY	
Name	cmp_topic
Activated	true
Method	btree
Unique	
Concurrent build	
If not exist	true
Only	true
Columns	
	topic
Expressions	
[1] value	
Include non-key columns	
With storage parameters	
[1] Fill factor	
Deduplicate items	
Tablespace	pg_default
Where constraint	
Comment	

2.1.2.2.4.4 Index

PROPERTY	
Name	cmp_start
Activated	true
Method	btree
Unique	
Concurrent build	
If not exist	true
Only	true
Columns	
	started_at
Expressions	
[1] value	
Include non-key columns	
With storage parameters	
[1] Fill factor	
Deduplicate items	
Tablespace	pg_default
Where constraint	
Comment	

2.1.2.2.5 campaigns Target Script

```
-- Generated by Hackolade Studio on: 2026-03-14T21:03:03+03:00
-- Application version: 8.9.6
-- PostgreSQL plugin version: 0.2.43
-- Custom scripts used: None

CREATE SCHEMA IF NOT EXISTS psql_scheme;
SET search_path TO psql_scheme, public;

CREATE TABLE IF NOT EXISTS psql_scheme.campaigns (
    campaign_id_sk bigserial PRIMARY KEY NOT NULL,
    campaign_id bigint NOT NULL,
    campaign_type varchar(32) NOT NULL,
    channel varchar(32) NOT NULL,
    topic varchar(255),
    started_at timestamp,
    finished_at timestamp,
    total_count bigint,
    subject_length integer,
    warmup_mode boolean DEFAULT false,
    subject_with_personalization boolean DEFAULT false,
    subject_with_deadline boolean DEFAULT false,
    subject_with_emoji boolean DEFAULT false,
    subject_with_bonuses boolean DEFAULT false,
    subject_with_discount boolean DEFAULT false,
    subject_with_saleout boolean DEFAULT false,
    CONSTRAINT campaigns_key UNIQUE (campaign_id, campaign_type)
) TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS cmp_id
ON ONLY psql_scheme.campaigns USING BTREE (campaign_id ASC, campaign_type ASC)
TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS cmp_channel
ON ONLY psql_scheme.campaigns USING BTREE (channel ASC)
TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS cmp_topic
ON ONLY psql_scheme.campaigns USING BTREE (topic ASC)
TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS cmp_start
ON ONLY psql_scheme.campaigns USING BTREE (started_at ASC)
TABLESPACE pg_default;
```

2.1.2.3 Table users

2.1.2.3.1 users Properties

PROPERTY	VALUE
Table	users
Technical name	users
Activated	true
Schema	psql_scheme
Additional properties	false
If not exists	true
Tablespace	pg_default

2.1.2.3.2 users Column

COLUMN	TYPE	REQ	KEY	DESCRIPTION	COMMENTS
user_id	bigint	true	pk, dk		

2.1.2.3.2.1 Column user_id

2.1.2.3.2.1.1 user_id properties

PROPERTY	VALUE
Business Name	user_id
Technical name	user_id
Activated	true
Type	numeric
Subtype	bigint
Not null	true
Primary key	true
Excl min	false
Excl max	false

2.1.2.3.3 users Composite keys

PROPERTY	VALUE
----------	-------

2.1.2.3.4 users Indexes

2.1.2.3.4.1 Index

PROPERTY	
Name	u_user
Activated	true
Method	btree
Unique	
Concurrent build	
If not exist	true
Only	true
Columns	
	user_id
Expressions	
[1] value	
Include non-key columns	
With storage parameters	
[1] Fill factor	
Deduplicate items	
Tablespace	pg_default
Where constraint	
Comment	

2.1.2.3.5 users Target Script

```
-- Generated by Hackolade Studio on: 2026-03-14T21:03:04+03:00
-- Application version: 8.9.6
-- PostgreSQL plugin version: 0.2.43
-- Custom scripts used: None
```

```
CREATE SCHEMA IF NOT EXISTS psql_scheme;
SET search_path TO psql_scheme, public;

CREATE TABLE IF NOT EXISTS psql_scheme.users (
    user_id bigint PRIMARY KEY NOT NULL
) TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS u_user
ON ONLY psql_scheme.users USING BTREE (user_id ASC)
TABLESPACE pg_default;
```

2.1.2.4 Table **campaigns_ab_test**

2.1.2.4.1 **campaigns_ab_test** Properties

PROPERTY	VALUE
Table	campaigns_ab_test
Technical name	campaigns_ab_test
Activated	true
Schema	psql_scheme
Additional properties	false
If not exists	true
Tablespace	pg_default

2.1.2.4.2 **campaigns_ab_test** Column

COLUMN	TYPE	REQ	KEY	DESCRIPTION	COMMENTS
campaign_id	bigint	true	fk		
ab_test	boolean	true			

2.1.2.4.2.1 Column **campaign_id**

2.1.2.4.2.1.1 **campaign_id** properties

PROPERTY	VALUE
Business Name	campaign_id
Technical name	campaign_id
Activated	true
Type	numeric
Subtype	bigint
Not null	true
Primary key	false
Unique	false
Foreign table	campaigns
Foreign field	campaign_id_sk
Relationship type	Foreign Key
Excl min	false
Excl max	false

2.1.2.4.2.2 Column **ab_test**

2.1.2.4.2.1 ab_test properties

PROPERTY	VALUE
Business Name	ab_test
Technical name	ab_test
Activated	true
Type	boolean
Not null	true

2.1.2.4.3 campaigns_ab_test Composite keys

PROPERTY	VALUE
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2.1.2.4.4 campaigns_ab_test Target Script

```
-- Generated by Hackolade Studio on: 2026-03-14T21:03:04+03:00
-- Application version: 8.9.6
-- PostgreSQL plugin version: 0.2.43
-- Custom scripts used: None
```

```
CREATE SCHEMA IF NOT EXISTS psql_scheme;
SET search_path TO psql_scheme, public;
```

```
CREATE TABLE IF NOT EXISTS psql_scheme.campaigns_ab_test (
    campaign_id bigint NOT NULL,
    ab_test boolean NOT NULL,
    CONSTRAINT fk_campaigns_id_to_campaigns_ab_test_campaign_id FOREIGN KEY (campaign_id) REFERENCES
    psql_scheme.campaigns (campaign_id_sk)
) TABLESPACE pg_default;
```

2.1.2.5 Table campaigns_hour_limit

2.1.2.5.1 campaigns_hour_limit Properties

PROPERTY	VALUE
Table	campaigns_hour_limit
Technical name	campaigns_hour_limit
Activated	true
Schema	psql_scheme
Additional properties	false
If not exists	true
Tablespace	pg_default

2.1.2.5.2 campaigns_hour_limit Column

COLUMN	TYPE	REQ	KEY	DESCRIPTION	COMMENTS
campaign_id	bigint	true	fk		
hour_limit	numeric(10, 2)	true			

2.1.2.5.2.1 Column campaign_id

2.1.2.5.2.1.1 campaign_id properties

PROPERTY	VALUE
Business Name	campaign_id
Technical name	campaign_id
Activated	true
Type	numeric
Subtype	bigint
Not null	true
Primary key	false
Unique	false
Foreign table	campaigns
Foreign field	campaign_id_sk
Relationship type	Foreign Key
Excl min	false
Excl max	false

2.1.2.5.2.2 Column hour_limit

2.1.2.5.2.2.1 hour_limit properties

PROPERTY	VALUE
Business Name	hour_limit
Technical name	hour_limit
Activated	true
Type	numeric
Subtype	numeric
Precision	10
Scale	2
Not null	true
Primary key	false
Unique	false
Excl min	false
Excl max	false

2.1.2.5.3 campaigns_hour_limit Composite keys

PROPERTY	VALUE
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2.1.2.5.4 campaigns_hour_limit Target Script

```
-- Generated by Hackolade Studio on: 2026-03-14T21:03:05+03:00
-- Application version: 8.9.6
-- PostgreSQL plugin version: 0.2.43
-- Custom scripts used: None

CREATE SCHEMA IF NOT EXISTS psql_scheme;
SET search_path TO psql_scheme, public;

CREATE TABLE IF NOT EXISTS psql_scheme.campaigns_hour_limit (
    campaign_id bigint NOT NULL,
    hour_limit numeric(10,2) NOT NULL,
    CONSTRAINT fk_campaigns_id_to_campaigns_ab_test_campaign_id FOREIGN KEY (campaign_id) REFERENCES
    psql_scheme.campaigns (campaign_id_sk)
) TABLESPACE pg_default;
```

2.1.2.6 Table **campaigns_test**

2.1.2.6.1 **campaigns_test** Properties

PROPERTY	VALUE
Table	campaigns_test
Technical name	campaigns_test
Activated	true
Schema	psql_scheme
Additional properties	false
If not exists	true
Tablespace	pg_default

2.1.2.6.2 **campaigns_test** Column

COLUMN	TYPE	REQ	KEY	DESCRIPTION	COMMENTS
campaign_id	bigint	true	fk		
is_test	boolean	true			

2.1.2.6.2.1 Column **campaign_id**

2.1.2.6.2.1.1 **campaign_id** properties

PROPERTY	VALUE
Business Name	campaign_id
Technical name	campaign_id
Activated	true
Type	numeric
Subtype	bigint
Not null	true
Primary key	false
Unique	false
Foreign table	campaigns
Foreign field	campaign_id_sk
Relationship type	Foreign Key
Excl min	false
Excl max	false

2.1.2.6.2.2 Column **is_test**

2.1.2.6.2.1 is_test properties

PROPERTY	VALUE
Business Name	is_test
Technical name	is_test
Activated	true
Type	boolean
Not null	true

2.1.2.6.3 campaigns_test Composite keys

PROPERTY	VALUE
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2.1.2.6.4 campaigns_test Target Script

```
-- Generated by Hackolade Studio on: 2026-03-14T21:03:05+03:00
-- Application version: 8.9.6
-- PostgreSQL plugin version: 0.2.43
-- Custom scripts used: None

CREATE SCHEMA IF NOT EXISTS psql_scheme;
SET search_path TO psql_scheme, public;

CREATE TABLE IF NOT EXISTS psql_scheme.campaigns_test (
  campaign_id bigint NOT NULL,
  is_test boolean NOT NULL,
  CONSTRAINT fk_campaigns_id_to_campaigns_ab_test_campaign_id FOREIGN KEY (campaign_id) REFERENCES
psql_scheme.campaigns (campaign_id_sk)
) TABLESPACE pg_default;
```

2.1.2.7 Table campaigns_position

2.1.2.7.1 campaigns_position Properties

PROPERTY	VALUE
Table	campaigns_position
Technical name	campaigns_position
Activated	true
Schema	psql_scheme
Additional properties	false
If not exists	true
Tablespace	pg_default

2.1.2.7.2 campaigns_position Column

COLUMN	TYPE	REQ	KEY	DESCRIPTION	COMMENTS
campaign_id	bigint	true	fk		
position	boolean	true			

2.1.2.7.2.1 Column campaign_id

2.1.2.7.2.1.1 campaign_id properties

PROPERTY	VALUE
Business Name	campaign_id
Technical name	campaign_id
Activated	true
Type	numeric
Subtype	bigint
Not null	true
Primary key	false
Unique	false
Foreign table	campaigns
Foreign field	campaign_id_sk
Relationship type	Foreign Key
Excl min	false
Excl max	false

2.1.2.7.2.2 Column position

2.1.2.7.2.2.1 position properties

PROPERTY	VALUE
Business Name	position
Technical name	position
Activated	true
Type	boolean
Not null	true

2.1.2.7.3 campaigns_position Composite keys

PROPERTY	VALUE
----------	-------

2.1.2.7.4 campaigns_position Target Script

```
-- Generated by Hackolade Studio on: 2026-03-14T21:03:06+03:00
-- Application version: 8.9.6
-- PostgreSQL plugin version: 0.2.43
-- Custom scripts used: None

CREATE SCHEMA IF NOT EXISTS psql_scheme;
SET search_path TO psql_scheme, public;

CREATE TABLE IF NOT EXISTS psql_scheme.campaigns_position (
    campaign_id bigint NOT NULL,
    position boolean NOT NULL,
    CONSTRAINT fk_campaigns_id_to_campaigns_ab_test_campaign_id FOREIGN KEY (campaign_id) REFERENCES
    psql_scheme.campaigns (campaign_id_sk)
) TABLESPACE pg_default;
```

2.1.2.8 Table messages

2.1.2.8.1 messages Properties

PROPERTY	VALUE
Table	messages
Technical name	messages
Activated	true
Schema	psql_scheme
Additional properties	false
If not exists	true
Tablespace	pg_default

2.1.2.8.2 messages Column

COLUMN	TYPE	REQ	KEY	DESCRIPTION	COMMENTS
id	bigserial	true	pk, dk		
message_id	uuid	true			
campaign_id	bigint	true	fk		
user_id	bigint	true	fk		
message_type	varchar(32)	true			
channel	varchar(32)	true			
date	date	true			
sent_at	timestamp	true			
created_at	timestamp	true			
updated_at	timestamp	true			
is_opened	boolean	false			
is_clicked	boolean	false			
is_unsubscribed	boolean	false			
is_hard_bounced	boolean	false			
is_soft_bounced	boolean	false			
is_complained	boolean	false			
is_blocked	boolean	false			
is_purchased	boolean	false			

2.1.2.8.2.1 Column id

2.1.2.8.2.1.1 id properties

PROPERTY	VALUE
Business Name	id
Technical name	id
Activated	true
Type	numeric
Subtype	bigserial
Not null	true
Primary key	true
Excl min	false
Excl max	false

2.1.2.8.2.2 Column message_id

2.1.2.8.2.2.1 **message_id** properties

PROPERTY	VALUE
Business Name	message_id
Technical name	message_id
Activated	true
Type	uuid
Not null	true
Unique	false

2.1.2.8.2.3 Column **campaign_id**

2.1.2.8.2.3.1 **campaign_id** properties

PROPERTY	VALUE
Business Name	campaign_id
Technical name	campaign_id
Activated	true
Type	numeric
Subtype	bigint
Not null	true
Primary key	false
Unique	false
Foreign table	campaigns
Foreign field	campaign_id_sk
Relationship type	Foreign Key
Excl min	false
Excl max	false

2.1.2.8.2.4 Column **user_id**

2.1.2.8.2.4.1 **user_id** properties

PROPERTY	VALUE
Business Name	user_id
Technical name	user_id
Activated	true
Type	numeric
Subtype	bigint
Not null	true
Primary key	false
Unique	false
Foreign table	users
Foreign field	user_id
Relationship type	Foreign Key
Excl min	false
Excl max	false

2.1.2.8.2.5 Column **message_type**

2.1.2.8.2.5.1 message_type properties

PROPERTY	VALUE
Business Name	message_type
Technical name	message_type
Activated	true
Type	char
Subtype	varchar
Length	32
Not null	true
Primary key	false
Unique	false

2.1.2.8.2.6 Column channel**2.1.2.8.2.6.1 channel properties**

PROPERTY	VALUE
Business Name	channel
Technical name	channel
Activated	true
Type	char
Subtype	varchar
Length	32
Not null	true
Primary key	false
Unique	false

2.1.2.8.2.7 Column date**2.1.2.8.2.7.1 date properties**

PROPERTY	VALUE
Business Name	date
Technical name	date
Activated	true
Type	datetime
Subtype	date
Not null	true
Primary key	false
Unique	false

2.1.2.8.2.8 Column sent_at

2.1.2.8.2.8.1 sent_at properties

PROPERTY	VALUE
Business Name	sent_at
Technical name	sent_at
Activated	true
Type	datetime
Subtype	timestamp
Not null	true
Primary key	false
Unique	false

2.1.2.8.2.9 Column created_at**2.1.2.8.2.9.1 created_at properties**

PROPERTY	VALUE
Business Name	created_at
Technical name	created_at
Activated	true
Type	datetime
Subtype	timestamp
Not null	true
Primary key	false
Unique	false

2.1.2.8.2.10 Column updated_at**2.1.2.8.2.10.1 updated_at properties**

PROPERTY	VALUE
Business Name	updated_at
Technical name	updated_at
Activated	true
Type	datetime
Subtype	timestamp
Not null	true
Primary key	false
Unique	false

2.1.2.8.2.11 Column is_opened**2.1.2.8.2.11.1 is_opened properties**

PROPERTY	VALUE
Business Name	is_opened
Technical name	is_opened
Activated	true
Type	boolean
Not null	false
Default	false

2.1.2.8.2.12 Column is_clicked

2.1.2.8.2.12.1 is_clicked properties

PROPERTY	VALUE
Business Name	is_clicked
Technical name	is_clicked
Activated	true
Type	boolean
Not null	false
Default	false

2.1.2.8.2.13 Column is_unsubscribed**2.1.2.8.2.13.1 is_unsubscribed properties**

PROPERTY	VALUE
Business Name	is_unsubscribed
Technical name	is_unsubscribed
Activated	true
Type	boolean
Not null	false
Default	false

2.1.2.8.2.14 Column is_hard_bounced**2.1.2.8.2.14.1 is_hard_bounced properties**

PROPERTY	VALUE
Business Name	is_hard_bounced
Technical name	is_hard_bounced
Activated	true
Type	boolean
Not null	false
Default	false

2.1.2.8.2.15 Column is_soft_bounced**2.1.2.8.2.15.1 is_soft_bounced properties**

PROPERTY	VALUE
Business Name	is_soft_bounced
Technical name	is_soft_bounced
Activated	true
Type	boolean
Not null	false
Default	false

2.1.2.8.2.16 Column is_complained

2.1.2.8.2.16.1 is_complained properties

PROPERTY	VALUE
Business Name	is_complained
Technical name	is_complained
Activated	true
Type	boolean
Not null	false
Default	false

2.1.2.8.2.17 Column is_blocked**2.1.2.8.2.17.1 is_blocked properties**

PROPERTY	VALUE
Business Name	is_blocked
Technical name	is_blocked
Activated	true
Type	boolean
Not null	false
Default	false

2.1.2.8.2.18 Column is_purchased**2.1.2.8.2.18.1 is_purchased properties**

PROPERTY	VALUE
Business Name	is_purchased
Technical name	is_purchased
Activated	true
Type	boolean
Not null	false
Default	false

2.1.2.8.3 messages Composite keys

PROPERTY	VALUE
----------	-------

2.1.2.8.4 messages Indexes**2.1.2.8.4.1 Index**

PROPERTY	
Name	ms_campaign
Activated	true
Method	btree
Unique	
Concurrent build	
If not exist	true
Only	true
Columns	
	campaign_id
Expressions	
[1] value	
Include non-key columns	
With storage parameters	
[1] Fill factor	
Deduplicate items	
Tablespace	pg_default
Where constraint	
Comment	

2.1.2.8.4.2 Index

PROPERTY	
Name	ms_user
Activated	true
Method	btree
Unique	
Concurrent build	
If not exist	true
Only	true
Columns	
	user_id
Expressions	
[1] value	
Include non-key columns	
With storage parameters	
[1] Fill factor	
Deduplicate items	
Tablespace	pg_default
Where constraint	
Comment	

2.1.2.8.4.3 Index

PROPERTY	
Name	ms_channel
Activated	true
Method	btree
Unique	
Concurrent build	
If not exist	true
Only	true
Columns	
	channel
Expressions	
[1] value	
Include non-key columns	
With storage parameters	
[1] Fill factor	
Deduplicate items	
Tablespace	pg_default
Where constraint	
Comment	

2.1.2.8.4.4 Index

PROPERTY	
Name	ms_date
Activated	true
Method	btree
Unique	
Concurrent build	
If not exist	true
Only	true
Columns	
	date
Expressions	
[1] value	
Include non-key columns	
With storage parameters	
[1] Fill factor	
Deduplicate items	
Tablespace	pg_default
Where constraint	
Comment	

2.1.2.8.4.5 Index

PROPERTY	
Name	ms_purchased
Activated	true
Method	btree
Unique	
Concurrent build	
If not exist	true
Only	true
Columns	
	is_purchased
Expressions	
[1] value	
Include non-key columns	
With storage parameters	
[1] Fill factor	
Deduplicate items	
Tablespace	pg_default
Where constraint	WHERE TRUE
Comment	

2.1.2.8.5 messages Target Script

```
-- Generated by Hackolade Studio on: 2026-03-14T21:03:06+03:00
-- Application version: 8.9.6
-- PostgreSQL plugin version: 0.2.43
-- Custom scripts used: None
```

```
CREATE SCHEMA IF NOT EXISTS psql_scheme;
SET search_path TO psql_scheme, public;
```

```
CREATE TABLE IF NOT EXISTS psql_scheme.messages (
    id bigserial PRIMARY KEY NOT NULL,
    message_id uuid NOT NULL,
    campaign_id bigint NOT NULL,
    user_id bigint NOT NULL,
    message_type varchar(32) NOT NULL,
    channel varchar(32) NOT NULL,
    date date NOT NULL,
    sent_at timestamp NOT NULL,
    created_at timestamp NOT NULL,
    updated_at timestamp NOT NULL,
    is_opened boolean DEFAULT false,
    is_clicked boolean DEFAULT false,
    is_unsubscribed boolean DEFAULT false,
    is_hard_bounced boolean DEFAULT false,
    is_soft_bounced boolean DEFAULT false,
    is_complained boolean DEFAULT false,
    is_blocked boolean DEFAULT false,
    is_purchased boolean DEFAULT false,
    CONSTRAINT fk_campaigns_campaign_id_sk_to_messages_campaign_id FOREIGN KEY (campaign_id) REFERENCES
psql_scheme.campaigns (campaign_id_sk),
    CONSTRAINT fk_users_user_id_to_messages_user_id FOREIGN KEY (user_id) REFERENCES psql_scheme.users
(user_id)
) TABLESPACE pg_default;
```

```
CREATE INDEX IF NOT EXISTS ms_campaign
ON ONLY psql_scheme.messages USING BTREE (campaign_id ASC)
TABLESPACE pg_default;
```

```
CREATE INDEX IF NOT EXISTS ms_user
ON ONLY psql_scheme.messages USING BTREE (user_id ASC)
TABLESPACE pg_default;
```

```
CREATE INDEX IF NOT EXISTS ms_channel
ON ONLY psql_scheme.messages USING BTREE (channel ASC)
TABLESPACE pg_default;
```

```
CREATE INDEX IF NOT EXISTS ms_date
ON ONLY psql_scheme.messages USING BTREE (date ASC)
TABLESPACE pg_default;
```

```
CREATE INDEX IF NOT EXISTS ms_purchased
ON ONLY psql_scheme.messages USING BTREE (is_purchased ASC)
TABLESPACE pg_default
WHERE WHERE TRUE;
```

2.1.2.9 Table **messages_device_info**

2.1.2.9.1 **messages_device_info** Properties

PROPERTY	VALUE
Table	messages_device_info
Technical name	messages_device_info
Activated	true
Schema	psql_scheme
Additional properties	false
If not exists	true
Tablespace	pg_default

2.1.2.9.2 messages_device_info Column

COLUMN	TYPE	REQ	KEY	DESCRIPTION	COMMENTS
message_id	bigint	true	fk		
stream	varchar(32)	true			
platform	varchar(32)	false			

2.1.2.9.2.1 Column message_id

2.1.2.9.2.1.1 message_id properties

PROPERTY	VALUE
Business Name	message_id
Technical name	message_id
Activated	true
Type	numeric
Subtype	bigint
Not null	true
Primary key	false
Unique	false
Foreign table	messages
Foreign field	id
Relationship type	Foreign Key
Excl min	false
Excl max	false

2.1.2.9.2.2 Column stream

2.1.2.9.2.2.1 stream properties

PROPERTY	VALUE
Business Name	stream
Technical name	stream
Activated	true
Type	char
Subtype	varchar
Length	32
Not null	true
Primary key	false
Unique	false

2.1.2.9.2.3 Column platform

2.1.2.9.2.3.1 platform properties

PROPERTY	VALUE
Business Name	platform
Technical name	platform
Activated	true
Type	char
Subtype	varchar
Length	32
Not null	false
Primary key	false
Unique	false

2.1.2.9.3 messages_device_info Composite keys

PROPERTY	VALUE
----------	-------

2.1.2.9.4 messages_device_info Target Script

```
-- Generated by Hackolade Studio on: 2026-03-14T21:03:06+03:00
-- Application version: 8.9.6
-- PostgreSQL plugin version: 0.2.43
-- Custom scripts used: None
```

```
CREATE SCHEMA IF NOT EXISTS psql_scheme;
SET search_path TO psql_scheme, public;
```

```
CREATE TABLE IF NOT EXISTS psql_scheme.messages_device_info (
  message_id bigint NOT NULL,
  stream varchar(32) NOT NULL,
  platform varchar(32),
  CONSTRAINT fk_messages_id_to_message_device_info_message_id FOREIGN KEY (message_id) REFERENCES
psql_scheme.messages (id)
) TABLESPACE pg_default;
```

2.1.2.10 Table messages_category

2.1.2.10.1 messages_category Properties

PROPERTY	VALUE
Table	messages_category
Technical name	messages_category
Activated	true
Schema	psql_scheme
Additional properties	false
If not exists	true
Tablespace	pg_default

2.1.2.10.2 messages_category Column

COLUMN	TYPE	REQ	KEY	DESCRIPTION	COMMENTS
message_id	bigint	true	fk		
category	varchar(128)	true			

2.1.2.10.2.1 Column message_id

2.1.2.10.2.1.1 message_id properties

PROPERTY	VALUE
Business Name	message_id
Technical name	message_id
Activated	true
Type	numeric
Subtype	bigint
Not null	true
Primary key	false
Unique	false
Foreign table	messages
Foreign field	id
Relationship type	Foreign Key
Excl min	false
Excl max	false

2.1.2.10.2.2 Column category

2.1.2.10.2.2.1 category properties

PROPERTY	VALUE
Business Name	category
Technical name	category
Activated	true
Type	char
Subtype	varchar
Length	128
Not null	true
Primary key	false
Unique	false

2.1.2.10.3 messages_category Composite keys

PROPERTY	VALUE
----------	-------

2.1.2.10.4 messages_category Target Script

```
-- Generated by Hackolade Studio on: 2026-03-14T21:03:07+03:00
-- Application version: 8.9.6
-- PostgreSQL plugin version: 0.2.43
-- Custom scripts used: None

CREATE SCHEMA IF NOT EXISTS psql_scheme;
SET search_path TO psql_scheme, public;

CREATE TABLE IF NOT EXISTS psql_scheme.messages_category (
    message_id bigint NOT NULL,
    category varchar(128) NOT NULL,
    CONSTRAINT fk_messages_id_to_messages_category_message_id FOREIGN KEY (message_id) REFERENCES
    psql_scheme.messages (id)
) TABLESPACE pg_default;
```

2.1.2.11 Table messages_recipient_info

2.1.2.11.1 messages_recipient_info Properties

PROPERTY	VALUE
Table	messages_recipient_info
Technical name	messages_recipient_info
Activated	true
Schema	psql_scheme
Additional properties	false
If not exists	true
Tablespace	pg_default

2.1.2.11.2 messages_recipient_info Column

COLUMN	TYPE	REQ	KEY	DESCRIPTION	COMMENTS
message_id	bigint	true	fk		
client_id	bigint	true	fk		
email_provider_id	bigint	false	fk		

2.1.2.11.2.1 Column message_id

2.1.2.11.2.1.1 message_id properties

PROPERTY	VALUE
Business Name	message_id
Technical name	message_id
Activated	true
Type	numeric
Subtype	bigint
Not null	true
Primary key	false
Unique	false
Foreign table	messages
Foreign field	id
Relationship type	Foreign Key
Excl min	false
Excl max	false

2.1.2.11.2.2 Column client_id

2.1.2.11.2.2.1 **client_id** properties

PROPERTY	VALUE
Business Name	client_id
Technical name	client_id
Activated	true
Type	numeric
Subtype	bigint
Not null	true
Primary key	false
Unique	false
Foreign table	clients
Foreign field	client_id
Relationship type	Foreign Key
Excl min	false
Excl max	false

2.1.2.11.2.3 Column **email_provider_id**

2.1.2.11.2.3.1 **email_provider_id** properties

PROPERTY	VALUE
Business Name	email_provider_id
Technical name	email_provider_id
Activated	true
Type	numeric
Subtype	bigint
Not null	false
Primary key	false
Unique	false
Foreign table	email_providers
Foreign field	id
Relationship type	Foreign Key
Excl min	false
Excl max	false

2.1.2.11.3 **messages_recipient_info** Composite keys

PROPERTY	VALUE
----------	-------

2.1.2.11.4 **messages_recipient_info** Target Script

```
-- Generated by Hackolade Studio on: 2026-03-14T21:03:07+03:00
-- Application version: 8.9.6
-- PostgreSQL plugin version: 0.2.43
-- Custom scripts used: None

CREATE SCHEMA IF NOT EXISTS psql_scheme;
SET search_path TO psql_scheme, public;

CREATE TABLE IF NOT EXISTS psql_scheme.messages_recipient_info (
    message_id bigint NOT NULL,
    client_id bigint NOT NULL,
    email_provider_id bigint,
    CONSTRAINT fk_messages_id_to_messages_recipient_info_message_id FOREIGN KEY (message_id) REFERENCES
psql_scheme.messages (id),
    CONSTRAINT fk_clients_client_id_to_messages_recipient_info_client_id FOREIGN KEY (client_id) REFERENCES
psql_scheme.clients (client_id),
    CONSTRAINT fk_email_providers_id_to_messages_recipient_info_email_provider_id FOREIGN KEY (email_provider_id)
REFERENCES psql_scheme.email_providers (id)
) TABLESPACE pg_default;
```

2.1.2.12 Table email_providers

2.1.2.12.1 email_providers Properties

PROPERTY	VALUE
Table	email_providers
Technical name	email_providers
Activated	true
Schema	psql_scheme
Additional properties	false
If not exists	true
Tablespace	pg_default

2.1.2.12.2 email_providers Column

COLUMN	TYPE	REQ	KEY	DESCRIPTION	COMMENTS
id	bigserial	true	pk, dk		
email_provider	varchar(255)	true			

2.1.2.12.2.1 Column id

2.1.2.12.2.1.1 id properties

PROPERTY	VALUE
Business Name	id
Technical name	id
Activated	true
Type	numeric
Subtype	bigserial
Generated column	false
Not null	true
Primary key	true
Excl min	false
Excl max	false

2.1.2.12.2.2 Column email_provider

2.1.2.12.2.1 email_provider properties

PROPERTY	VALUE
Business Name	email_provider
Technical name	email_provider
Activated	true
Type	char
Subtype	varchar
Length	255
Not null	true
Primary key	false
Unique	false

2.1.2.12.3 email_providers Composite keys

PROPERTY	VALUE
----------	-------

2.1.2.12.4 email_providers Target Script

```
-- Generated by Hackolade Studio on: 2026-03-14T21:03:08+03:00
-- Application version: 8.9.6
-- PostgreSQL plugin version: 0.2.43
-- Custom scripts used: None

CREATE SCHEMA IF NOT EXISTS psql_scheme;
SET search_path TO psql_scheme, public;

CREATE TABLE IF NOT EXISTS psql_scheme.email_providers (
  id bigserial PRIMARY KEY NOT NULL,
  email_provider varchar(255) NOT NULL
) TABLESPACE pg_default;
```

2.1.2.13 Table message_opens

2.1.2.13.1 message_opens Properties

PROPERTY	VALUE
Table	message_opens
Technical name	message_opens
Activated	true
Schema	psql_scheme
Additional properties	false
If not exists	true
Tablespace	pg_default

2.1.2.13.2 message_opens Column

COLUMN	TYPE	REQ	KEY	DESCRIPTION	COMMENTS
message_id	bigint	true	fk		
opened_first_time_at	timestamp	true			
opened_last_time_at	timestamp	true			

2.1.2.13.2.1 Column message_id

2.1.2.13.2.1.1 message_id properties

PROPERTY	VALUE
Business Name	message_id
Technical name	message_id
Activated	true
Type	numeric
Subtype	bigint
Not null	true
Primary key	false
Unique	false
Foreign table	messages
Foreign field	id
Relationship type	Foreign Key
Excl min	false
Excl max	false

2.1.2.13.2.2 Column opened_first_time_at**2.1.2.13.2.2.1 opened_first_time_at properties**

PROPERTY	VALUE
Business Name	opened_first_time_at
Technical name	opened_first_time_at
Activated	true
Type	datetime
Subtype	timestamp
Not null	true
Primary key	false
Unique	false

2.1.2.13.2.3 Column opened_last_time_at**2.1.2.13.2.3.1 opened_last_time_at properties**

PROPERTY	VALUE
Business Name	opened_last_time_at
Technical name	opened_last_time_at
Activated	true
Type	datetime
Subtype	timestamp
Not null	true
Primary key	false
Unique	false

2.1.2.13.3 message_opens Composite keys

PROPERTY	VALUE
----------	-------

2.1.2.13.4 message_opens Target Script

```
-- Generated by Hackolade Studio on: 2026-03-14T21:03:08+03:00
-- Application version: 8.9.6
-- PostgreSQL plugin version: 0.2.43
-- Custom scripts used: None

CREATE SCHEMA IF NOT EXISTS psql_scheme;
SET search_path TO psql_scheme, public;

CREATE TABLE IF NOT EXISTS psql_scheme.message_opens (
    message_id bigint NOT NULL,
    opened_first_time_at timestamp NOT NULL,
    opened_last_time_at timestamp NOT NULL,
    CONSTRAINT fk_messages_id_to_message_opens_message_id FOREIGN KEY (message_id) REFERENCES
    psql_scheme.messages (id)
) TABLESPACE pg_default;
```

2.1.2.14 Table **message_clicks**

2.1.2.14.1 **message_clicks** Properties

PROPERTY	VALUE
Table	message_clicks
Technical name	message_clicks
Activated	true
Schema	psql_scheme
Additional properties	false
If not exists	true
Tablespace	pg_default

2.1.2.14.2 **message_clicks** Column

COLUMN	TYPE	REQ	KEY	DESCRIPTION	COMMENTS
message_id	bigint	true	fk		
clicked_first_time_at	timestamp	true			
clicked_last_time_at	timestamp	true			

2.1.2.14.2.1 Column **message_id**

2.1.2.14.2.1.1 **message_id** properties

PROPERTY	VALUE
Business Name	message_id
Technical name	message_id
Activated	true
Type	numeric
Subtype	bigint
Not null	true
Primary key	false
Unique	false
Foreign table	messages
Foreign field	id
Relationship type	Foreign Key
Excl min	false
Excl max	false

2.1.2.14.2.2 Column **clicked_first_time_at**

2.1.2.14.2.2.1 clicked_first_time_at properties

PROPERTY	VALUE
Business Name	clicked_first_time_at
Technical name	clicked_first_time_at
Activated	true
Type	datetime
Subtype	timestamp
Not null	true
Primary key	false
Unique	false

2.1.2.14.2.3 Column clicked_last_time_at

2.1.2.14.2.3.1 clicked_last_time_at properties

PROPERTY	VALUE
Business Name	clicked_last_time_at
Technical name	clicked_last_time_at
Activated	true
Type	datetime
Subtype	timestamp
Not null	true
Primary key	false
Unique	false

2.1.2.14.3 message_clicks Composite keys

PROPERTY	VALUE
----------	-------

2.1.2.14.4 message_clicks Target Script

```
-- Generated by Hackolade Studio on: 2026-03-14T21:03:09+03:00
-- Application version: 8.9.6
-- PostgreSQL plugin version: 0.2.43
-- Custom scripts used: None

CREATE SCHEMA IF NOT EXISTS psql_scheme;
SET search_path TO psql_scheme, public;

CREATE TABLE IF NOT EXISTS psql_scheme.message_clicks (
    message_id bigint NOT NULL,
    clicked_first_time_at timestamp NOT NULL,
    clicked_last_time_at timestamp NOT NULL,
    CONSTRAINT fk_messages_id_to_message_opens_message_id FOREIGN KEY (message_id) REFERENCES
psql_scheme.messages (id)
) TABLESPACE pg_default;
```

2.1.2.15 Table message_negatives

2.1.2.15.1 message_negatives Properties

PROPERTY	VALUE
Table	message_negatives
Technical name	message_negatives
Activated	true
Schema	psql_scheme
Additional properties	false
If not exists	true
Tablespace	pg_default

2.1.2.15.2 message_negatives Column

COLUMN	TYPE	REQ	KEY	DESCRIPTION	COMMENTS
message_id	bigint	true	fk		
unsubscribed_at	timestamp	false			
hard_bounced_at	timestamp	false			
soft_bounced_at	timestamp	false			
complained_at	timestamp	false			
blocked_at	timestamp	false			

2.1.2.15.2.1 Column message_id

2.1.2.15.2.1.1 message_id properties

PROPERTY	VALUE
Business Name	message_id
Technical name	message_id
Activated	true
Type	numeric
Subtype	bigint
Not null	true
Primary key	false
Unique	false
Foreign table	messages
Foreign field	id
Relationship type	Foreign Key
Excl min	false
Excl max	false

2.1.2.15.2.2 Column unsubscribed_at

2.1.2.15.2.2.1 unsubscribed_at properties

PROPERTY	VALUE
Business Name	unsubscribed_at
Technical name	unsubscribed_at
Activated	true
Type	datetime
Subtype	timestamp
Not null	false
Primary key	false
Unique	false

2.1.2.15.2.3 Column hard_bounced_at

2.1.2.15.2.3.1 **hard_bounced_at** properties

PROPERTY	VALUE
Business Name	hard_bounced_at
Technical name	hard_bounced_at
Activated	true
Type	datetime
Subtype	timestamp
Not null	false
Primary key	false
Unique	false

2.1.2.15.2.4 Column **soft_bounced_at**

2.1.2.15.2.4.1 **soft_bounced_at** properties

PROPERTY	VALUE
Business Name	soft_bounced_at
Technical name	soft_bounced_at
Activated	true
Type	datetime
Subtype	timestamp
Not null	false
Primary key	false
Unique	false

2.1.2.15.2.5 Column **complained_at**

2.1.2.15.2.5.1 **complained_at** properties

PROPERTY	VALUE
Business Name	complained_at
Technical name	complained_at
Activated	true
Type	datetime
Subtype	timestamp
Not null	false
Primary key	false
Unique	false

2.1.2.15.2.6 Column **blocked_at**

2.1.2.15.2.6.1 **blocked_at** properties

PROPERTY	VALUE
Business Name	blocked_at
Technical name	blocked_at
Activated	true
Type	datetime
Subtype	timestamp
Not null	false
Primary key	false
Unique	false

2.1.2.15.3 **message_negatives** Composite keys

PROPERTY	VALUE
----------	-------

2.1.2.15.4 message_negatives Target Script

```
-- Generated by Hackolade Studio on: 2026-03-14T21:03:09+03:00
-- Application version: 8.9.6
-- PostgreSQL plugin version: 0.2.43
-- Custom scripts used: None

CREATE SCHEMA IF NOT EXISTS psql_scheme;
SET search_path TO psql_scheme, public;

CREATE TABLE IF NOT EXISTS psql_scheme.message_negatives (
    message_id bigint NOT NULL,
    unsubscribed_at timestamp,
    hard_bounced_at timestamp,
    soft_bounced_at timestamp,
    complained_at timestamp,
    blocked_at timestamp,
    CONSTRAINT fk_messages_id_to_message_opens_message_id FOREIGN KEY (message_id) REFERENCES
    psql_scheme.messages (id)
) TABLESPACE pg_default;
```

2.1.2.16 Table message_purchases

2.1.2.16.1 message_purchases Properties

PROPERTY	VALUE
Table	message_purchases
Technical name	message_purchases
Activated	true
Schema	psql_scheme
Additional properties	false
If not exists	true
Tablespace	pg_default

2.1.2.16.2 message_purchases Column

COLUMN	TYPE	REQ	KEY	DESCRIPTION	COMMENTS
message_id	bigint	true	fk		
purchased_at	timestamp	true			

2.1.2.16.2.1 Column message_id

2.1.2.16.2.1.1 message_id properties

PROPERTY	VALUE
Business Name	message_id
Technical name	message_id
Activated	true
Type	numeric
Subtype	bigint
Not null	true
Primary key	false
Unique	false
Foreign table	messages
Foreign field	id
Relationship type	Foreign Key
Excl min	false
Excl max	false

2.1.2.16.2.2 Column purchased_at

2.1.2.16.2.2.1 purchased_at properties

PROPERTY	VALUE
Business Name	purchased_at
Technical name	purchased_at
Activated	true
Type	datetime
Subtype	timestamp
Not null	true
Primary key	false
Unique	false

2.1.2.16.3 message_purchases Composite keys

PROPERTY	VALUE
----------	-------

2.1.2.16.4 message_purchases Target Script

```
-- Generated by Hackolade Studio on: 2026-03-14T21:03:10+03:00
-- Application version: 8.9.6
-- PostgreSQL plugin version: 0.2.43
-- Custom scripts used: None

CREATE SCHEMA IF NOT EXISTS psql_scheme;
SET search_path TO psql_scheme, public;

CREATE TABLE IF NOT EXISTS psql_scheme.message_purchases (
  message_id bigint NOT NULL,
  purchased_at timestamp NOT NULL,
  CONSTRAINT fk_messages_id_to_message_opens_message_id FOREIGN KEY (message_id) REFERENCES
psql_scheme.messages (id)
) TABLESPACE pg_default;
```

2.1.2.17 Table events

2.1.2.17.1 events Properties

PROPERTY	VALUE
Table	events
Technical name	events
Activated	true
Schema	psql_scheme
Additional properties	false
If not exists	true
Tablespace	pg_default

2.1.2.17.2 events Column

COLUMN	TYPE	REQ	KEY	DESCRIPTION	COMMENTS
id	bigserial	true	pk		
event_time	timestamp	true			
event_type	varchar(32)	true			
product_id	bigint	true	fk		
category_id	bigint	true	fk		
price	numeric(10, 2)	true			
user_id	bigint	true	fk		
user_session	uuid	true			

2.1.2.17.2.1 Column id

2.1.2.17.2.1.1 id properties

PROPERTY	VALUE
Business Name	id
Technical name	id
Activated	true
Type	numeric
Subtype	bigserial
Not null	true
Primary key	true
Excl min	false
Excl max	false

2.1.2.17.2.2 Column event_time

2.1.2.17.2.2.1 event_time properties

PROPERTY	VALUE
Business Name	event_time
Technical name	event_time
Activated	true
Type	datetime
Subtype	timestamp
Not null	true
Primary key	false
Unique	false

2.1.2.17.2.3 Column event_type

2.1.2.17.2.3.1 event_type properties

PROPERTY	VALUE
Business Name	event_type
Technical name	event_type
Activated	true
Type	char
Subtype	varchar
Length	32
Not null	true
Primary key	false
Unique	false

2.1.2.17.2.4 Column product_id**2.1.2.17.2.4.1 product_id properties**

PROPERTY	VALUE
Business Name	product_id
Technical name	product_id
Activated	true
Type	numeric
Subtype	bigint
Not null	true
Primary key	false
Unique	false
Foreign table	products
Foreign field	product_id
Relationship type	Foreign Key
Excl min	false
Excl max	false

2.1.2.17.2.5 Column category_id**2.1.2.17.2.5.1 category_id properties**

PROPERTY	VALUE
Business Name	category_id
Technical name	category_id
Activated	true
Type	numeric
Subtype	bigint
Not null	true
Primary key	false
Unique	false
Foreign table	categories
Foreign field	category_id
Relationship type	Foreign Key
Excl min	false
Excl max	false

2.1.2.17.2.6 Column price

2.1.2.17.2.6.1 price properties

PROPERTY	VALUE
Business Name	price
Technical name	price
Activated	true
Type	numeric
Subtype	numeric
Precision	10
Scale	2
Not null	true
Primary key	false
Unique	false
Excl min	false
Excl max	false

2.1.2.17.2.7 Column user_id**2.1.2.17.2.7.1 user_id properties**

PROPERTY	VALUE
Business Name	user_id
Technical name	user_id
Activated	true
Type	numeric
Subtype	bigint
Not null	true
Primary key	false
Unique	false
Foreign table	users
Foreign field	user_id
Relationship type	Foreign Key
Excl min	false
Excl max	false

2.1.2.17.2.8 Column user_session**2.1.2.17.2.8.1 user_session properties**

PROPERTY	VALUE
Business Name	user_session
Technical name	user_session
Activated	true
Type	uuid
Not null	true
Primary key	false
Unique	false

2.1.2.17.3 events Composite keys

PROPERTY	VALUE
----------	-------

2.1.2.17.4 events Indexes**2.1.2.17.4.1 Index**

PROPERTY	
Name	e_time
Activated	true
Method	btree
Unique	
Concurrent build	
If not exist	true
Only	true
Columns	
	event_time
Expressions	
[1] value	
Include non-key columns	
With storage parameters	
[1] Fill factor	
Deduplicate items	
Tablespace	pg_default
Where constraint	
Comment	

2.1.2.17.4.2 Index

PROPERTY	
Name	e_product
Activated	true
Method	btree
Unique	
Concurrent build	
If not exist	true
Only	true
Columns	
	product_id
Expressions	
[1] value	
Include non-key columns	
With storage parameters	
[1] Fill factor	
Deduplicate items	
Tablespace	pg_default
Where constraint	
Comment	

2.1.2.17.4.3 Index

PROPERTY	
Name	e_user
Activated	true
Method	btree
Unique	
Concurrent build	
If not exist	true
Only	true
Columns	
	user_id
Expressions	
[1] value	
Include non-key columns	
With storage parameters	
[1] Fill factor	
Deduplicate items	
Tablespace	pg_default
Where constraint	
Comment	

2.1.2.17.4.4 Index

PROPERTY	
Name	e_type
Activated	true
Method	btree
Unique	
Concurrent build	
If not exist	true
Only	true
Columns	
	event_type
Expressions	
[1] value	
Include non-key columns	
With storage parameters	
[1] Fill factor	
Deduplicate items	
Tablespace	pg_default
Where constraint	WHERE 'purchase'
Comment	

2.1.2.17.4.5 Index

PROPERTY	
Name	e_category
Activated	true
Method	btree
Unique	
Concurrent build	
If not exist	true
Only	true
Columns	
	category_id
Expressions	
[1] value	
Include non-key columns	
With storage parameters	
[1] Fill factor	
Deduplicate items	
Tablespace	pg_default
Where constraint	
Comment	

2.1.2.17.5 events Target Script

```
-- Generated by Hackolade Studio on: 2026-03-14T21:03:10+03:00
-- Application version: 8.9.6
-- PostgreSQL plugin version: 0.2.43
-- Custom scripts used: None

CREATE SCHEMA IF NOT EXISTS psql_scheme;
SET search_path TO psql_scheme, public;

CREATE TABLE IF NOT EXISTS psql_scheme.events (
  id bigserial PRIMARY KEY NOT NULL,
  event_time timestamp NOT NULL,
  event_type varchar(32) NOT NULL,
  product_id bigint NOT NULL,
  category_id bigint NOT NULL,
  price numeric(10,2) NOT NULL,
  user_id bigint NOT NULL,
  user_session uuid NOT NULL,
  CONSTRAINT fk_users_user_id_to_events_user_id FOREIGN KEY (user_id) REFERENCES psql_scheme.users (user_id),
  CONSTRAINT fk_products_product_id_to_events_product_id FOREIGN KEY (product_id) REFERENCES
psql_scheme.products (product_id),
  CONSTRAINT fk_categories_category_id_to_events_category_id FOREIGN KEY (category_id) REFERENCES
psql_scheme.categories (category_id)
) TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS e_time
ON ONLY psql_scheme.events USING BTREE (event_time ASC)
TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS e_product
ON ONLY psql_scheme.events USING BTREE (product_id ASC)
TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS e_user
ON ONLY psql_scheme.events USING BTREE (user_id ASC)
TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS e_type
ON ONLY psql_scheme.events USING BTREE (event_type ASC)
TABLESPACE pg_default
WHERE WHERE 'purchase';

CREATE INDEX IF NOT EXISTS e_categoty
ON ONLY psql_scheme.events USING BTREE (category_id ASC)
TABLESPACE pg_default;
```

2.1.2.18 Table products

2.1.2.18.1 products Properties

PROPERTY	VALUE
Table	products
Technical name	products
Activated	true
Schema	psql_scheme
Additional properties	false
If not exists	true
Tablespace	pg_default

2.1.2.18.2 products Column

COLUMN	TYPE	REQ	KEY	DESCRIPTION	COMMENTS
product_id	bigint	true	pk, dk		
brand	varchar(128)	false			

2.1.2.18.2.1 Column product_id

2.1.2.18.2.1.1 product_id properties

PROPERTY	VALUE
Business Name	product_id
Technical name	product_id
Activated	true
Type	numeric
Subtype	bigint
Not null	true
Primary key	true
Excl min	false
Excl max	false

2.1.2.18.2.2 Column brand

2.1.2.18.2.2.1 brand properties

PROPERTY	VALUE
Business Name	brand
Technical name	brand
Activated	true
Type	char
Subtype	varchar
Length	128
Not null	false
Primary key	false
Unique	false

2.1.2.18.3 products Composite keys

PROPERTY	VALUE
----------	-------

2.1.2.18.4 products Indexes

2.1.2.18.4.1 Index

PROPERTY	
Name	p_brand
Activated	true
Method	btree
Unique	
Concurrent build	
If not exist	true
Only	true
Columns	
	brand
Expressions	
[1] value	
Include non-key columns	
With storage parameters	
[1] Fill factor	
Deduplicate items	
Tablespace	pg_default
Where constraint	
Comment	

2.1.2.18.5 products Target Script

```
-- Generated by Hackolade Studio on: 2026-03-14T21:03:11+03:00
-- Application version: 8.9.6
-- PostgreSQL plugin version: 0.2.43
-- Custom scripts used: None

CREATE SCHEMA IF NOT EXISTS psql_scheme;
SET search_path TO psql_scheme, public;

CREATE TABLE IF NOT EXISTS psql_scheme.products (
  product_id bigint PRIMARY KEY NOT NULL,
  brand varchar(128)
) TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS p_brand
ON ONLY psql_scheme.products USING BTREE (brand ASC)
TABLESPACE pg_default;
```

2.1.2.19 Table categories

2.1.2.19.1 categories Properties

PROPERTY	VALUE
Table	categories
Technical name	categories
Activated	true
Schema	psql_scheme
Additional properties	false
If not exists	true
Tablespace	pg_default

2.1.2.19.2 categories Column

COLUMN	TYPE	REQ	KEY	DESCRIPTION	COMMENTS
category_id	bigint	true	pk, dk		
category_code	text	false			

2.1.2.19.2.1 Column **category_id**

2.1.2.19.2.1.1 **category_id** properties

PROPERTY	VALUE
Business Name	category_id
Technical name	category_id
Activated	true
Type	numeric
Subtype	bigint
Not null	true
Primary key	true
Excl min	false
Excl max	false

2.1.2.19.2.2 Column **category_code**

2.1.2.19.2.2.1 **category_code** properties

PROPERTY	VALUE
Business Name	category_code
Technical name	category_code
Activated	true
Type	char
Subtype	text
Not null	false
Primary key	false
Unique	false

2.1.2.19.3 **categories** Composite keys

PROPERTY	VALUE
----------	-------

2.1.2.19.4 **categories** Indexes

2.1.2.19.4.1 **Index**

PROPERTY	
Name	c_cat_code
Activated	true
Method	btree
Unique	
Concurrent build	
If not exist	true
Only	true
Columns	
	category_code
Expressions	
[1] value	
Include non-key columns	
With storage parameters	
[1] Fill factor	
Deduplicate items	
Tablespace	pg_default
Where constraint	
Comment	

2.1.2.19.5 categories Target Script

```
-- Generated by Hackolade Studio on: 2026-03-14T21:03:11+03:00
-- Application version: 8.9.6
-- PostgreSQL plugin version: 0.2.43
-- Custom scripts used: None

CREATE SCHEMA IF NOT EXISTS psql_scheme;
SET search_path TO psql_scheme, public;

CREATE TABLE IF NOT EXISTS psql_scheme.categories (
    category_id bigint PRIMARY KEY NOT NULL,
    category_code text
) TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS c_cat_code
ON ONLY psql_scheme.categories USING BTREE (category_code ASC)
TABLESPACE pg_default;
```

2.1.2.20 Table friends

2.1.2.20.1 friends Properties

PROPERTY	VALUE
Table	friends
Technical name	friends
Activated	true
Schema	psql_scheme
Additional properties	false
If not exists	true
Tablespace	pg_default

2.1.2.20.2 friends Column

COLUMN	TYPE	REQ	KEY	DESCRIPTION	COMMENTS
user_id	bigint	true	pk, fk		
friend_id	bigint	true	pk, fk		

2.1.2.20.2.1 Column **user_id**

2.1.2.20.2.1.1 **user_id** properties

PROPERTY	VALUE
Business Name	user_id
Technical name	user_id
Activated	true
Type	numeric
Subtype	bigint
Not null	true
Primary key	true
Unique	false
Foreign table	users
Foreign field	user_id
Relationship type	Foreign Key
Excl min	false
Excl max	false

2.1.2.20.2.2 Column **friend_id**

2.1.2.20.2.2.1 **friend_id** properties

PROPERTY	VALUE
Business Name	friend_id
Technical name	friend_id
Activated	true
Type	numeric
Subtype	bigint
Not null	true
Primary key	true
Unique	false
Foreign table	users
Foreign field	user_id
Relationship type	Foreign Key
Excl min	false
Excl max	false

2.1.2.20.3 **friends** Composite keys

PROPERTY	VALUE
Primary key	
[1] Constraint name	user_friend_key
Key	
	user_id
	friend_id
Include non-key columns	
With storage parameters	
Index tablespace	
Comment	
Deferrable	

2.1.2.20.4 friends Indexes

2.1.2.20.4.1 Index

PROPERTY	
Name	f_user
Activated	true
Method	btree
Unique	
Concurrent build	
If not exist	true
Only	true
Columns	
	user_id
Expressions	
[1] value	
Include non-key columns	
With storage parameters	
[1] Fill factor	
Deduplicate items	
Tablespace	pg_default
Where constraint	
Comment	

2.1.2.20.4.2 Index

PROPERTY	
Name	f_friend
Activated	true
Method	btree
Unique	
Concurrent build	
If not exist	true
Only	true
Columns	
	friend_id
Expressions	
[1] value	
Include non-key columns	
With storage parameters	
[1] Fill factor	
Deduplicate items	
Tablespace	pg_default
Where constraint	
Comment	

2.1.2.20.5 friends Target Script

```
-- Generated by Hackolade Studio on: 2026-03-14T21:03:12+03:00
-- Application version: 8.9.6
-- PostgreSQL plugin version: 0.2.43
-- Custom scripts used: None

CREATE SCHEMA IF NOT EXISTS psql_scheme;
SET search_path TO psql_scheme, public;

CREATE TABLE IF NOT EXISTS psql_scheme.friends (
    user_id bigint NOT NULL,
    friend_id bigint NOT NULL,
    CONSTRAINT user_friend_key PRIMARY KEY (user_id, friend_id),
    CONSTRAINT fk_users_user_id_to_friends_user_id FOREIGN KEY (user_id) REFERENCES psql_scheme.users (user_id),
    CONSTRAINT fk_users_user_id_to_friends_friend_id FOREIGN KEY (friend_id) REFERENCES psql_scheme.users (user_id)
) TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS f_user
ON ONLY psql_scheme.friends USING BTREE (user_id ASC)
TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS f_friend
ON ONLY psql_scheme.friends USING BTREE (friend_id ASC)
TABLESPACE pg_default;
```

2.1.3 psql_scheme Target Script

```
-- Generated by Hackolade Studio on: 2026-03-14T21:03:12+03:00
-- Application version: 8.9.6
-- PostgreSQL plugin version: 0.2.43
-- Custom scripts used: None
```

```
CREATE SCHEMA IF NOT EXISTS psql_scheme;
SET search_path TO psql_scheme, public;
```

```
CREATE TABLE IF NOT EXISTS psql_scheme.users (
    user_id bigint PRIMARY KEY NOT NULL
) TABLESPACE pg_default;
```

```
CREATE TABLE IF NOT EXISTS psql_scheme.clients (
    client_id bigint PRIMARY KEY NOT NULL,
    user_id bigint NOT NULL,
    user_device_id integer NOT NULL,
    first_purchase_date date,
    CONSTRAINT fk_users_user_id_to_clients_user_id FOREIGN KEY (user_id) REFERENCES psql_scheme.users (user_id)
) TABLESPACE pg_default;
```

```
CREATE TABLE IF NOT EXISTS psql_scheme.campaigns (
    campaign_id_sk bigserial PRIMARY KEY NOT NULL,
    campaign_id bigint NOT NULL,
    campaign_type varchar(32) NOT NULL,
    channel varchar(32) NOT NULL,
    topic varchar(255),
    started_at timestamp,
    finished_at timestamp,
    total_count bigint,
    subject_length integer,
    warmup_mode boolean DEFAULT false,
    subject_with_personalization boolean DEFAULT false,
    subject_with_deadline boolean DEFAULT false,
    subject_with_emoji boolean DEFAULT false,
    subject_with_bonuses boolean DEFAULT false,
    subject_with_discount boolean DEFAULT false,
    subject_with_saleout boolean DEFAULT false,
    CONSTRAINT campaigns_key UNIQUE (campaign_id, campaign_type)
) TABLESPACE pg_default;
```

```
CREATE TABLE IF NOT EXISTS psql_scheme.campaigns_ab_test (
    campaign_id bigint NOT NULL,
    ab_test boolean NOT NULL,
    CONSTRAINT fk_campaigns_id_to_campaigns_ab_test_campaign_id FOREIGN KEY (campaign_id) REFERENCES
psql_scheme.campaigns (campaign_id_sk)
) TABLESPACE pg_default;
```

```
CREATE TABLE IF NOT EXISTS psql_scheme.campaigns_hour_limit (
    campaign_id bigint NOT NULL,
    hour_limit numeric(10,2) NOT NULL,
    CONSTRAINT fk_campaigns_id_to_campaigns_ab_test_campaign_id FOREIGN KEY (campaign_id) REFERENCES
psql_scheme.campaigns (campaign_id_sk)
) TABLESPACE pg_default;
```

```
CREATE TABLE IF NOT EXISTS psql_scheme.campaigns_test (
    campaign_id bigint NOT NULL,
    is_test boolean NOT NULL,
    CONSTRAINT fk_campaigns_id_to_campaigns_ab_test_campaign_id FOREIGN KEY (campaign_id) REFERENCES
psql_scheme.campaigns (campaign_id_sk)
) TABLESPACE pg_default;
```

```
CREATE TABLE IF NOT EXISTS psql_scheme.campaigns_position (  
    campaign_id bigint NOT NULL,  
    position boolean NOT NULL,  
    CONSTRAINT fk_campaigns_id_to_campaigns_ab_test_campaign_id FOREIGN KEY (campaign_id) REFERENCES  
psql_scheme.campaigns (campaign_id_sk)  
) TABLESPACE pg_default;  
  
CREATE TABLE IF NOT EXISTS psql_scheme.messages (  
    id bigserial PRIMARY KEY NOT NULL,  
    message_id uuid NOT NULL,  
    campaign_id bigint NOT NULL,  
    user_id bigint NOT NULL,  
    message_type varchar(32) NOT NULL,  
    channel varchar(32) NOT NULL,  
    date date NOT NULL,  
    sent_at timestamp NOT NULL,  
    created_at timestamp NOT NULL,  
    updated_at timestamp NOT NULL,  
    is_opened boolean DEFAULT false,  
    is_clicked boolean DEFAULT false,  
    is_unsubscribed boolean DEFAULT false,  
    is_hard_bounced boolean DEFAULT false,  
    is_soft_bounced boolean DEFAULT false,  
    is_complained boolean DEFAULT false,  
    is_blocked boolean DEFAULT false,  
    is_purchased boolean DEFAULT false,  
    CONSTRAINT fk_campaigns_campaign_id_sk_to_messages_campaign_id FOREIGN KEY (campaign_id) REFERENCES  
psql_scheme.campaigns (campaign_id_sk),  
    CONSTRAINT fk_users_user_id_to_messages_user_id FOREIGN KEY (user_id) REFERENCES psql_scheme.users  
(user_id)  
) TABLESPACE pg_default;  
  
CREATE TABLE IF NOT EXISTS psql_scheme.messages_device_info (  
    message_id bigint NOT NULL,  
    stream varchar(32) NOT NULL,  
    platform varchar(32),  
    CONSTRAINT fk_messages_id_to_message_device_info_message_id FOREIGN KEY (message_id) REFERENCES  
psql_scheme.messages (id)  
) TABLESPACE pg_default;  
  
CREATE TABLE IF NOT EXISTS psql_scheme.messages_category (  
    message_id bigint NOT NULL,  
    category varchar(128) NOT NULL,  
    CONSTRAINT fk_messages_id_to_messages_category_message_id FOREIGN KEY (message_id) REFERENCES  
psql_scheme.messages (id)  
) TABLESPACE pg_default;  
  
CREATE TABLE IF NOT EXISTS psql_scheme.email_providers (  
    id bigserial PRIMARY KEY NOT NULL,  
    email_provider varchar(255) NOT NULL  
) TABLESPACE pg_default;  
  
CREATE TABLE IF NOT EXISTS psql_scheme.messages_recipient_info (  
    message_id bigint NOT NULL,  
    client_id bigint NOT NULL,  
    email_provider_id bigint,  
    CONSTRAINT fk_messages_id_to_messages_recipient_info_message_id FOREIGN KEY (message_id) REFERENCES  
psql_scheme.messages (id),  
    CONSTRAINT fk_clients_client_id_to_messages_recipient_info_client_id FOREIGN KEY (client_id) REFERENCES  
psql_scheme.clients (client_id),
```

```
CONSTRAINT fk_email_providers_id_to_messages_recipient_info_email_provider_id FOREIGN KEY (email_provider_id)
REFERENCES psql_scheme.email_providers (id)
) TABLESPACE pg_default;

CREATE TABLE IF NOT EXISTS psql_scheme.message_opens (
    message_id bigint NOT NULL,
    opened_first_time_at timestamp NOT NULL,
    opened_last_time_at timestamp NOT NULL,
    CONSTRAINT fk_messages_id_to_message_opens_message_id FOREIGN KEY (message_id) REFERENCES
psql_scheme.messages (id)
) TABLESPACE pg_default;

CREATE TABLE IF NOT EXISTS psql_scheme.message_clicks (
    message_id bigint NOT NULL,
    clicked_first_time_at timestamp NOT NULL,
    clicked_last_time_at timestamp NOT NULL,
    CONSTRAINT fk_messages_id_to_message_opens_message_id FOREIGN KEY (message_id) REFERENCES
psql_scheme.messages (id)
) TABLESPACE pg_default;

CREATE TABLE IF NOT EXISTS psql_scheme.message_negatives (
    message_id bigint NOT NULL,
    unsubscribed_at timestamp,
    hard_bounced_at timestamp,
    soft_bounced_at timestamp,
    complained_at timestamp,
    blocked_at timestamp,
    CONSTRAINT fk_messages_id_to_message_opens_message_id FOREIGN KEY (message_id) REFERENCES
psql_scheme.messages (id)
) TABLESPACE pg_default;

CREATE TABLE IF NOT EXISTS psql_scheme.message_purchases (
    message_id bigint NOT NULL,
    purchased_at timestamp NOT NULL,
    CONSTRAINT fk_messages_id_to_message_opens_message_id FOREIGN KEY (message_id) REFERENCES
psql_scheme.messages (id)
) TABLESPACE pg_default;

CREATE TABLE IF NOT EXISTS psql_scheme.products (
    product_id bigint PRIMARY KEY NOT NULL,
    brand varchar(128)
) TABLESPACE pg_default;

CREATE TABLE IF NOT EXISTS psql_scheme.categories (
    category_id bigint PRIMARY KEY NOT NULL,
    category_code text
) TABLESPACE pg_default;

CREATE TABLE IF NOT EXISTS psql_scheme.events (
    id bigserial PRIMARY KEY NOT NULL,
    event_time timestamp NOT NULL,
    event_type varchar(32) NOT NULL,
    product_id bigint NOT NULL,
    category_id bigint NOT NULL,
    price numeric(10,2) NOT NULL,
    user_id bigint NOT NULL,
    user_session uuid NOT NULL,
    CONSTRAINT fk_users_user_id_to_events_user_id FOREIGN KEY (user_id) REFERENCES psql_scheme.users (user_id),
    CONSTRAINT fk_products_product_id_to_events_product_id FOREIGN KEY (product_id) REFERENCES
psql_scheme.products (product_id),
    CONSTRAINT fk_categories_category_id_to_events_category_id FOREIGN KEY (category_id) REFERENCES
```

```
psql_scheme.categories (category_id)
) TABLESPACE pg_default;

CREATE TABLE IF NOT EXISTS psql_scheme.friends (
    user_id bigint NOT NULL,
    friend_id bigint NOT NULL,
    CONSTRAINT user_friend_key PRIMARY KEY (user_id, friend_id),
    CONSTRAINT fk_users_user_id_to_friends_user_id FOREIGN KEY (user_id) REFERENCES psql_scheme.users (user_id),
    CONSTRAINT fk_users_user_id_to_friends_friend_id FOREIGN KEY (friend_id) REFERENCES psql_scheme.users (user_id)
) TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS u_user
ON ONLY psql_scheme.users USING BTREE (user_id ASC)
TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS "user"
ON ONLY psql_scheme.clients USING BTREE (first_purchase_date ASC)
TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS cmp_id
ON ONLY psql_scheme.campaigns USING BTREE (campaign_id ASC, campaign_type ASC)
TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS cmp_channel
ON ONLY psql_scheme.campaigns USING BTREE (channel ASC)
TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS cmp_topic
ON ONLY psql_scheme.campaigns USING BTREE (topic ASC)
TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS cmp_start
ON ONLY psql_scheme.campaigns USING BTREE (started_at ASC)
TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS ms_campaign
ON ONLY psql_scheme.messages USING BTREE (campaign_id ASC)
TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS ms_user
ON ONLY psql_scheme.messages USING BTREE (user_id ASC)
TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS ms_channel
ON ONLY psql_scheme.messages USING BTREE (channel ASC)
TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS ms_date
ON ONLY psql_scheme.messages USING BTREE (date ASC)
TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS ms_purchased
ON ONLY psql_scheme.messages USING BTREE (is_purchased ASC)
TABLESPACE pg_default
WHERE WHERE TRUE;

CREATE INDEX IF NOT EXISTS p_brand
ON ONLY psql_scheme.products USING BTREE (brand ASC)
TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS c_cat_code
```

```
ON ONLY psql_scheme.categories USING BTREE (category_code ASC)
TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS e_time
ON ONLY psql_scheme.events USING BTREE (event_time ASC)
TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS e_product
ON ONLY psql_scheme.events USING BTREE (product_id ASC)
TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS e_user
ON ONLY psql_scheme.events USING BTREE (user_id ASC)
TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS e_type
ON ONLY psql_scheme.events USING BTREE (event_type ASC)
TABLESPACE pg_default
WHERE WHERE 'purchase';

CREATE INDEX IF NOT EXISTS e_category
ON ONLY psql_scheme.events USING BTREE (category_id ASC)
TABLESPACE pg_default;

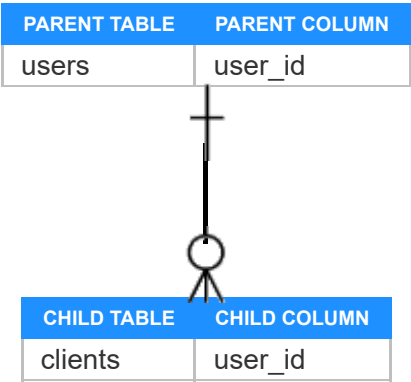
CREATE INDEX IF NOT EXISTS f_user
ON ONLY psql_scheme.friends USING BTREE (user_id ASC)
TABLESPACE pg_default;

CREATE INDEX IF NOT EXISTS f_friend
ON ONLY psql_scheme.friends USING BTREE (friend_id ASC)
TABLESPACE pg_default;
```

3. RELATIONSHIPS

3.1 Relationship **fk_users_user_id_to_clients_user_id**

3.1.1 **fk_users_user_id_to_clients_user_id** Diagram

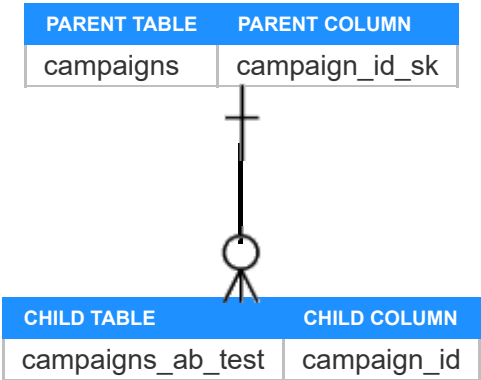


3.1.2 fk_users_user_id_to_clients_user_id Properties

PROPERTY	VALUE
Name	fk_users_user_id_to_clients_user_id
Activated	true
Relationship type	Foreign Key
Parent table	users
Parent field	user_id
Parent cardinality	1
Child table	clients
Child field	user_id
Child cardinality	0..n

3.2 Relationship fk_campaigns_id_to_campaigns_ab_test_campaign_id

3.2.1 fk_campaigns_id_to_campaigns_ab_test_campaign_id Diagram

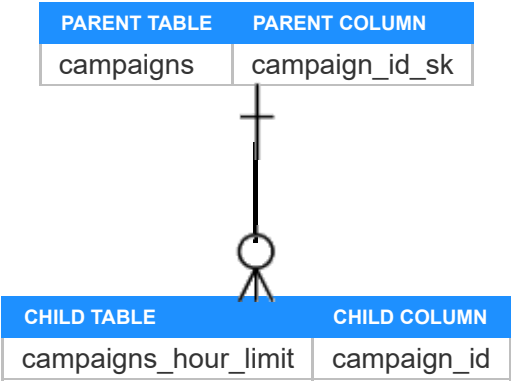


3.2.2 fk_campaigns_id_to_campaigns_ab_test_campaign_id Properties

PROPERTY	VALUE
Name	fk_campaigns_id_to_campaigns_ab_test_campaign_id
Activated	true
Relationship type	Foreign Key
Parent table	campaigns
Parent field	campaign_id_sk
Parent cardinality	1
Child table	campaigns_ab_test
Child field	campaign_id
Child cardinality	0..n

3.3 Relationship fk_campaigns_id_to_campaigns_ab_test_campaign_id

3.3.1 fk_campaigns_id_to_campaigns_ab_test_campaign_id Diagram

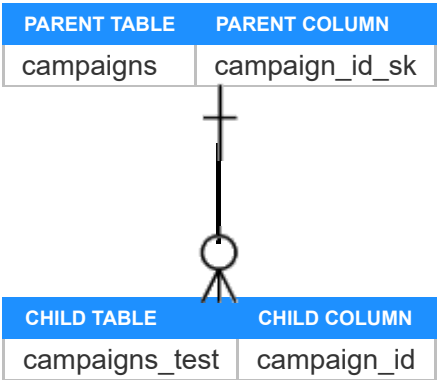


3.3.2 fk_campaigns_id_to_campaigns_ab_test_campaign_id Properties

PROPERTY	VALUE
Name	fk_campaigns_id_to_campaigns_ab_test_campaign_id
Activated	true
Relationship type	Foreign Key
Parent table	campaigns
Parent field	campaign_id_sk
Parent cardinality	1
Child table	campaigns_hour_limit
Child field	campaign_id
Child cardinality	0..n

3.4 Relationship fk_campaigns_id_to_campaigns_ab_test_campaign_id

3.4.1 fk_campaigns_id_to_campaigns_ab_test_campaign_id Diagram

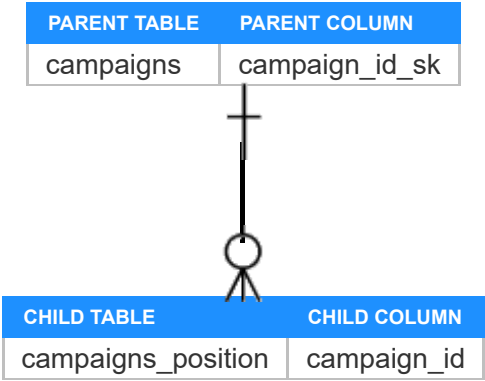


3.4.2 fk_campaigns_id_to_campaigns_ab_test_campaign_id Properties

PROPERTY	VALUE
Name	fk_campaigns_id_to_campaigns_ab_test_campaign_id
Activated	true
Relationship type	Foreign Key
Parent table	campaigns
Parent field	campaign_id_sk
Parent cardinality	1
Child table	campaigns_test
Child field	campaign_id
Child cardinality	0..n

3.5 Relationship fk_campaigns_id_to_campaigns_ab_test_campaign_id

3.5.1 fk_campaigns_id_to_campaigns_ab_test_campaign_id Diagram

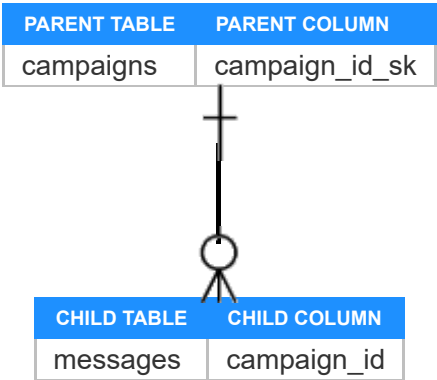


3.5.2 fk_campaigns_id_to_campaigns_ab_test_campaign_id Properties

PROPERTY	VALUE
Name	fk_campaigns_id_to_campaigns_ab_test_campaign_id
Activated	true
Relationship type	Foreign Key
Parent table	campaigns
Parent field	campaign_id_sk
Parent cardinality	1
Child table	campaigns_position
Child field	campaign_id
Child cardinality	0..n

3.6 Relationship fk_campaigns_campaign_id_sk_to_messages_campaign_id

3.6.1 fk_campaigns_campaign_id_sk_to_messages_campaign_id Diagram

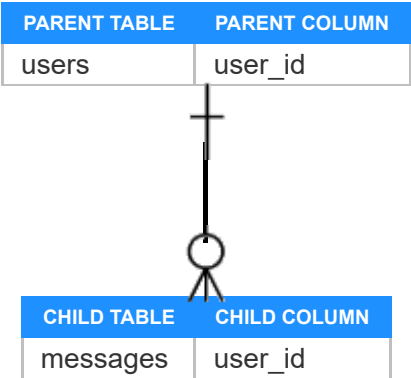


3.6.2 fk_campaigns_campaign_id_sk_to_messages_campaign_id Properties

PROPERTY	VALUE
Name	fk_campaigns_campaign_id_sk_to_messages_campaign_id
Activated	true
Relationship type	Foreign Key
Parent table	campaigns
Parent field	campaign_id_sk
Parent cardinality	1
Child table	messages
Child field	campaign_id
Child cardinality	0..n

3.7 Relationship fk_users_user_id_to_messages_user_id

3.7.1 fk_users_user_id_to_messages_user_id Diagram

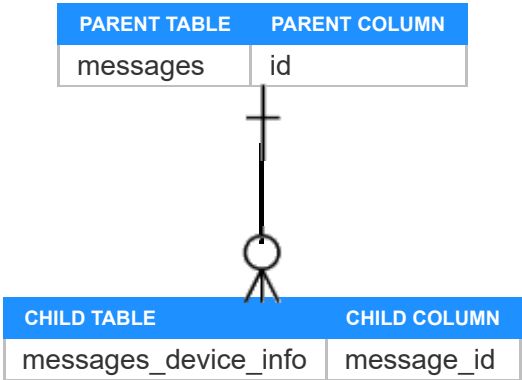


3.7.2 fk_users_user_id_to_messages_user_id Properties

PROPERTY	VALUE
Name	fk_users_user_id_to_messages_user_id
Activated	true
Relationship type	Foreign Key
Parent table	users
Parent field	user_id
Parent cardinality	1
Child table	messages
Child field	user_id
Child cardinality	0..n

3.8 Relationship fk_messages_id_to_message_device_info_message_id

3.8.1 fk_messages_id_to_message_device_info_message_id Diagram

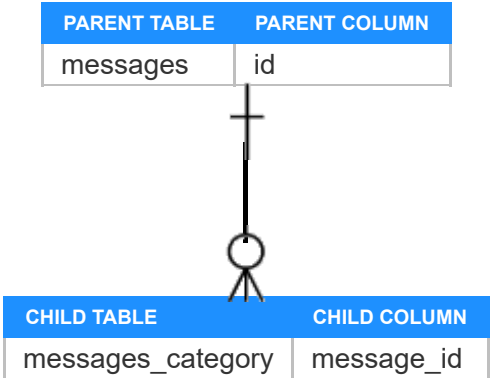


3.8.2 fk_messages_id_to_message_device_info_message_id Properties

PROPERTY	VALUE
Name	fk_messages_id_to_message_device_info_message_id
Activated	true
Relationship type	Foreign Key
Parent table	messages
Parent field	id
Parent cardinality	1
Child table	messages_device_info
Child field	message_id
Child cardinality	0..n

3.9 Relationship fk_messages_id_to_messages_category_message_id

3.9.1 fk_messages_id_to_messages_category_message_id Diagram

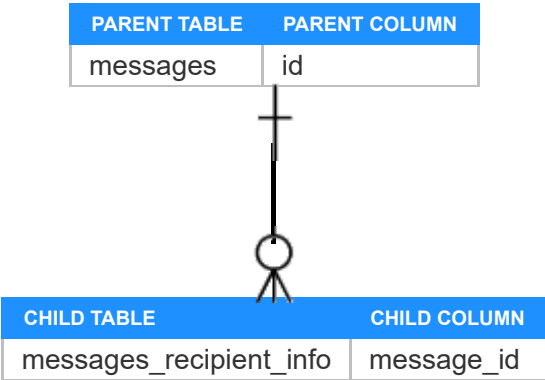


3.9.2 fk_messages_id_to_messages_category_message_id Properties

PROPERTY	VALUE
Name	fk_messages_id_to_messages_category_message_id
Activated	true
Relationship type	Foreign Key
Parent table	messages
Parent field	id
Parent cardinality	1
Child table	messages_category
Child field	message_id
Child cardinality	0..n

3.10 Relationship fk_messages_id_to_messages_recipient_info_message_id

3.10.1 fk_messages_id_to_messages_recipient_info_message_id Diagram

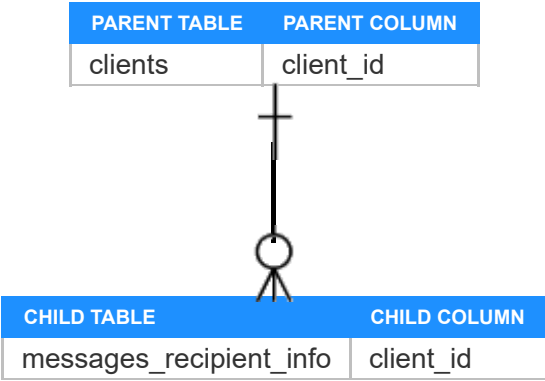


3.10.2 fk_messages_id_to_messages_recipient_info_message_id Properties

PROPERTY	VALUE
Name	fk_messages_id_to_messages_recipient_info_message_id
Activated	true
Relationship type	Foreign Key
Parent table	messages
Parent field	id
Parent cardinality	1
Child table	messages_recipient_info
Child field	message_id
Child cardinality	0..n

3.11 Relationship fk_clients_client_id_to_messages_recipient_info_client_id

3.11.1 fk_clients_client_id_to_messages_recipient_info_client_id Diagram

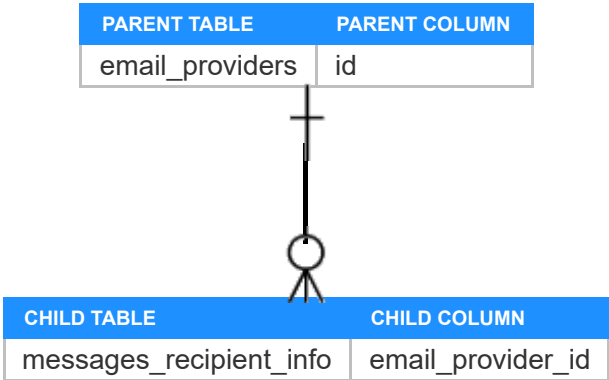


3.11.2 fk_clients_client_id_to_messages_recipient_info_client_id Properties

PROPERTY	VALUE
Name	fk_clients_client_id_to_messages_recipient_info_client_id
Activated	true
Relationship type	Foreign Key
Parent table	clients
Parent field	client_id
Parent cardinality	1
Child table	messages_recipient_info
Child field	client_id
Child cardinality	0..n

3.12 Relationship fk_email_providers_id_to_messages_recipient_info_email_provider_id

3.12.1 fk_email_providers_id_to_messages_recipient_info_email_provider_id Diagram

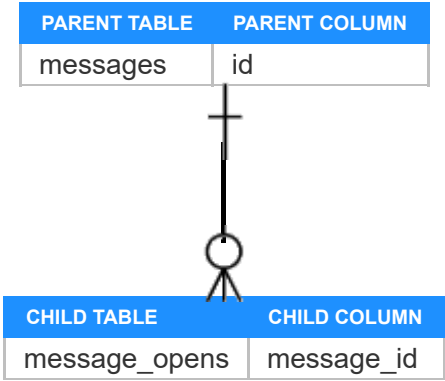


3.12.2 fk_email_providers_id_to_messages_recipient_info_email_provider_id Properties

PROPERTY	VALUE
Name	fk_email_providers_id_to_messages_recipient_info_email_provider_id
Activated	true
Relationship type	Foreign Key
Parent table	email_providers
Parent field	id
Parent cardinality	1
Child table	messages_recipient_info
Child field	email_provider_id
Child cardinality	0..n

3.13 Relationship fk_messages_id_to_message_opens_message_id

3.13.1 fk_messages_id_to_message_opens_message_id Diagram

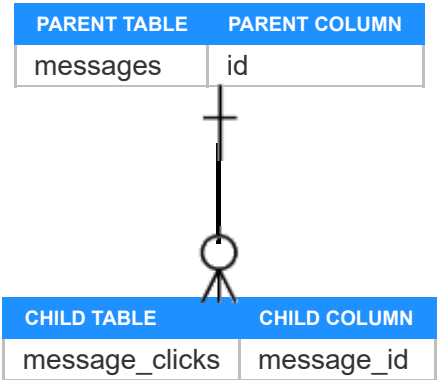


3.13.2 fk_messages_id_to_message_opens_message_id Properties

PROPERTY	VALUE
Name	fk_messages_id_to_message_opens_message_id
Activated	true
Relationship type	Foreign Key
Parent table	messages
Parent field	id
Parent cardinality	1
Child table	message_opens
Child field	message_id
Child cardinality	0..n

3.14 Relationship fk_messages_id_to_message_opens_message_id

3.14.1 fk_messages_id_to_message_opens_message_id Diagram

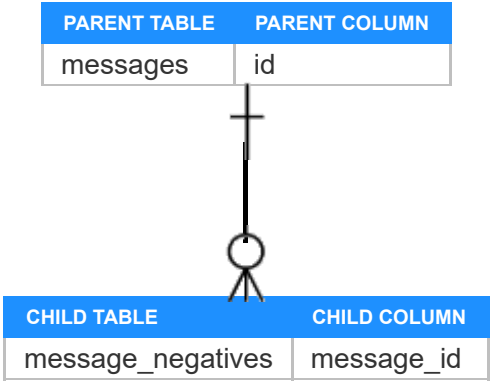


3.14.2 fk_messages_id_to_message_opens_message_id Properties

PROPERTY	VALUE
Name	fk_messages_id_to_message_opens_message_id
Activated	true
Relationship type	Foreign Key
Parent table	messages
Parent field	id
Parent cardinality	1
Child table	message_clicks
Child field	message_id
Child cardinality	0..n

3.15 Relationship fk_messages_id_to_message_opens_message_id

3.15.1 fk_messages_id_to_message_opens_message_id Diagram

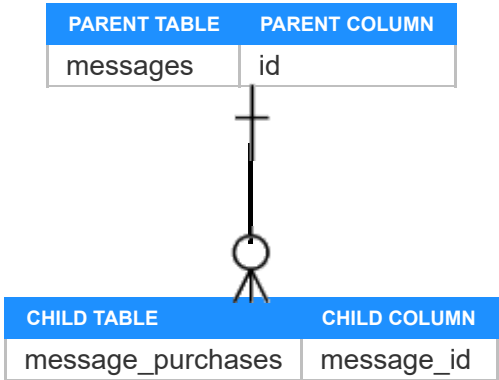


3.15.2 fk_messages_id_to_message_opens_message_id Properties

PROPERTY	VALUE
Name	fk_messages_id_to_message_opens_message_id
Activated	true
Relationship type	Foreign Key
Parent table	messages
Parent field	id
Parent cardinality	1
Child table	message_negatives
Child field	message_id
Child cardinality	0..n

3.16 Relationship fk_messages_id_to_message_opens_message_id

3.16.1 fk_messages_id_to_message_opens_message_id Diagram

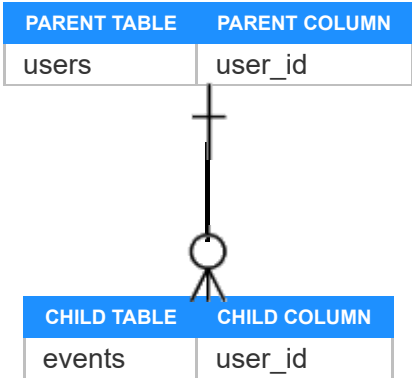


3.16.2 fk_messages_id_to_message_opens_message_id Properties

PROPERTY	VALUE
Name	fk_messages_id_to_message_opens_message_id
Activated	true
Relationship type	Foreign Key
Parent table	messages
Parent field	id
Parent cardinality	1
Child table	message_purchases
Child field	message_id
Child cardinality	0..n

3.17 Relationship fk_users_user_id_to_events_user_id

3.17.1 fk_users_user_id_to_events_user_id Diagram

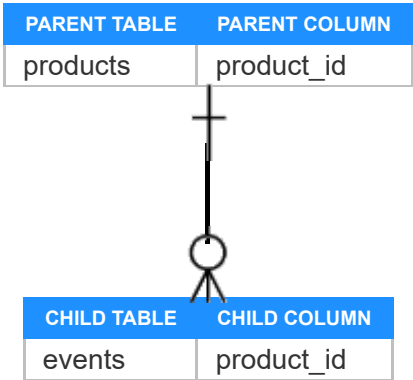


3.17.2 fk_users_user_id_to_events_user_id Properties

PROPERTY	VALUE
Name	fk_users_user_id_to_events_user_id
Activated	true
Relationship type	Foreign Key
Parent table	users
Parent field	user_id
Parent cardinality	1
Child table	events
Child field	user_id
Child cardinality	0..n

3.18 Relationship fk_products_product_id_to_events_product_id

3.18.1 fk_products_product_id_to_events_product_id Diagram

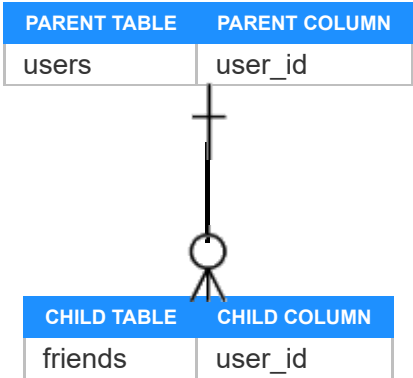


3.18.2 fk_products_product_id_to_events_product_id Properties

PROPERTY	VALUE
Name	fk_products_product_id_to_events_product_id
Activated	true
Relationship type	Foreign Key
Parent table	products
Parent field	product_id
Parent cardinality	1
Child table	events
Child field	product_id
Child cardinality	0..n

3.19 Relationship fk_users_user_id_to_friends_user_id

3.19.1 fk_users_user_id_to_friends_user_id Diagram

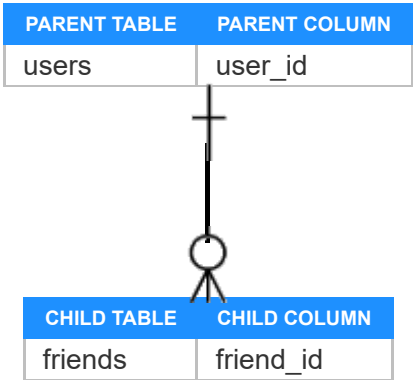


3.19.2 fk_users_user_id_to_friends_user_id Properties

PROPERTY	VALUE
Name	fk_users_user_id_to_friends_user_id
Activated	true
Relationship type	Foreign Key
Parent table	users
Parent field	user_id
Parent cardinality	1
Child table	friends
Child field	user_id
Child cardinality	0..n

3.20 Relationship fk_users_user_id_to_friends_friend_id

3.20.1 fk_users_user_id_to_friends_friend_id Diagram

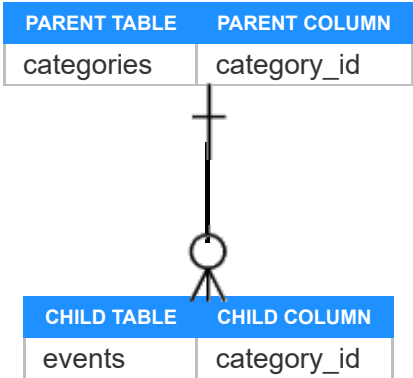


3.20.2 fk_users_user_id_to_friends_friend_id Properties

PROPERTY	VALUE
Name	fk_users_user_id_to_friends_friend_id
Activated	true
Relationship type	Foreign Key
Parent table	users
Parent field	user_id
Parent cardinality	1
Child table	friends
Child field	friend_id
Child cardinality	0..n

3.21 Relationship fk_categories_category_id_to_events_category_id

3.21.1 fk_categories_category_id_to_events_category_id Diagram



3.21.2 fk_categories_category_id_to_events_category_id Properties

PROPERTY	VALUE
Name	fk_categories_category_id_to_events_category_id
Activated	true
Relationship type	Foreign Key
Parent table	categories
Parent field	category_id
Parent cardinality	1
Child table	events
Child field	category_id
Child cardinality	0..n